Fate Contagion and Termination Criteria for the Juggler Map

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9 September 2026

Abstract

The Juggler map sends an even positive integer to the integer part of its square root and an odd positive integer to the integer part of its three-halves power. We prove that every nonempty set A closed under taking preimages satisfies $\sum_{n\in A,\, n\le x} 1/n \ge c(\log x)^{\lambda}$, for all sufficiently large x and every $0 < \lambda < \lambda^{**}$, where $\lambda^{**} \approx 0.4926$ is an explicitly specified root. Thus every realized cycle basin and the set of unbounded orbits, if nonempty, obey this lower bound. The proof combines exact inverse intervals, a monotone parity sweep, classical exponential-sum estimates, and a finite list of disjoint parity productions.

For any fixed $N_0 \ge 2$ such that every start in $[1, N_0]$ reaches 1, universal termination is equivalent to an eventual-entry statement: all but $O(y(\log y)^{-e})$ odd starts in $(y, 2y]$ enter $[1, N_0]$, for some $e > 1 - \lambda^{**}$. We give sufficient parity-cylinder and exponential-moment hypotheses at depth $O(\log\log y)$ for a stronger statement with a time bound. These hypotheses remain unproved. A first-letter decomposition identifies the contribution of failures beginning with two odd steps, and an abstract production model describes possible improvements of the exponent. A separate appendix gives a conditional exponent 0.5392. Selected combinatorial lemmas are formalized in Lean 4; the analytic estimates are mathematical arguments outside the formalization, and the numerical experiments are observations. Neither universal termination nor the exclusion of a nontrivial cycle or an unbounded orbit is established.

2020 Mathematics Subject Classification: 11B37, 11L07, 37P99.
Keywords: Juggler map; preimage sets; logarithmic counting; parity cylinders; integer dynamics.

1. Introduction

For a positive integer n, define

$$J(n) = \begin{cases} \lfloor \sqrt{n} \rfloor, & n \text{ even}, \\ \lfloor n^{3/2} \rfloor, & n \text{ odd}. \end{cases}$$

The Juggler sequence was introduced by Pickover [1] and is sequence A094683 of the OEIS [2]. Starting from any positive integer, the orbit under J appears always to reach 1; this has been verified for all starts up to $3.5 \cdot 10^8$ [11]. The map belongs to the family of floor-power maps, and its dynamics resembles the $3x + 1$ problem [3, 4] in one respect and differs from it in another. The resemblance: a parity decides whether the orbit goes up or down, and the heuristic model of the orbit is a random walk with negative drift, here on the scale $\log\log n$ with steps $+\log_2 3 - 1$ (odd) and -1 (even). The difference: the Juggler map moves by *powers*, so a single contracting certificate sends n to n^e with $e < 1$, and the one-step preimages of an integer are not a residue class but an interval.

This paper is about the second feature and what it forces. The narrative has four steps.

Contagion. Let R be the set of starts that reach 1, F its complement, $B(C)$ the basin of a nontrivial cycle C, and D the set of divergent starts. Each is *backward-closed*: if $J(n)$ belongs to it, so does n. Theorem 1 proves that every nonempty backward-closed set is large in logarithmic density, with an explicit exponent, by a downward recursion over two exact productions and the finite composites in Appendix D. Applied to the fate classes: if a single start fails to reach 1, the failures have logarithmic count $\gg (\log x)^{\lambda}$ and, on infinitely many dyadic blocks, natural density $\gg (\log y)^{\lambda-1}$, for every $0 < \lambda < \lambda^{**}$. This excludes no fate; it fixes the quantitative shape of the trichotomy.

Odd generation. A set that is closed both forwards and backwards and avoids 1 is the E-forest (iterated even preimages) over its odd members, and its odd members are the odd preimages of its intersection with the sparse set $S = \{\lfloor m^{3/2} \rfloor : m \text{ odd}\}$ (Theorem 2). An arbitrary failure set is thereby replaced by sparse odd-image seeds together with deterministic even ancestry. This is the geometry that the later sections use.

The almost-all equivalence. Tao [6] proved that almost all Collatz orbits (in logarithmic density) attain almost bounded values, with a target $f(N) \to \infty$ arbitrarily slowly. For the Juggler map, contagion and odd generation together turn a bounded-target statement into the conjecture: every positive integer reaches 1 if and only if all but $O(y(\log y)^{-e})$ odd starts in $(y, 2y]$ enter $[1, N_0]$, for some $e > 1 - \lambda^{**} \approx 0.5074$ (Theorem 3). The threshold is the complement of the contagion exponent. The Collatz preimage lower bound of Krasikov–Lagarias [7], discussed with its target restriction in Section 1.3, does not by itself yield this implication.

Sufficient estimates at growing depth. The exponent walk needs to fall by $L(y) = \log_2(\log(2y)/\log N_0)$ to put a start in $(y, 2y]$ below the floor. A Chernoff estimate counts words whose walk fails to cross this level by depth $\lceil CL(y) \rceil$. Turning this word count into a count of integer starts requires additional arithmetic information. Sections 8–9 give three sufficient forms of that information: upper bounds on bad cylinders, one-sided control of their continuations, and a bound on one exponential moment of live starts. A bound on tilted odd shares implies the moment bound. None is established here.

The first-letter decomposition in Section 6 relates the remaining OO-contribution to failures at a larger scale. Section 10 connects this term with live-start counts and states a uniformity requirement for a particular cylinder-counting method. These are reductions and limitations of the stated estimates, not necessity results for every possible proof of termination.

1.1 Main results

Write $\ell_A(u, v] = \sum_{n \in A,\, u < n \le v} 1/n$. Throughout, $N_0 \ge 2$ is any integer with $[1, N_0] \subseteq R$ (the Lean-verified value is 260; the certified computational value is $3.5 \cdot 10^8$ [11]).

Theorem 1 (fate contagion). Let $A \subseteq \mathbb{N}$ be nonempty and backward-closed, and let $0 < \lambda < \lambda^{**} \approx 0.4926$, the root of

$$2^{-\lambda} + \tfrac{1}{9}(\tfrac{3}{8})^{\lambda} + \tfrac{2}{9}(\tfrac{3}{4})^{\lambda} + \sum_{k=2}^{6} 3^{-(k+1)} \left(\tfrac{1}{2}(\tfrac{3}{4})^{k}\right)^{\lambda} = 1. \tag{1.1}$$

(The V_5 truncation of the first seven terms is 0.4924; the V_4 truncation of the first six terms is 0.4916; the V_3 truncation of the first five is 0.4891; the $OEOEE$ truncation of the first four is 0.4801; the pairing-only root of the first three is 0.4480.) There are $c > 0$ and x_0, depending on A and λ, with

$$\sum_{\substack{n \in A \\ n \le x}} \frac{1}{n} \;\ge\; c\,(\log x)^{\lambda} \qquad (x \ge x_0).$$

Consequently each fate class realized by at least one start — R, the basin of any existing nontrivial cycle, the divergent set — satisfies this bound, and on infinitely many dyadic blocks has natural density $\gg (\log y)^{\lambda-1}$. (Theorem 5.3, Corollaries 5.4, 5.5.)

The finite productions in Section 5.7 and Appendix D give this exponent. Their formal infinite series has limiting root 0.4927, but only the stated finite truncation is used here. The ideal characteristic model in Proposition 5.12 has root 1; the other run-model values, including 0.8414, are conditional calculations and are not attained density exponents.

Theorem 2 (odd generation). Let A be forward- and backward-closed with $1 \notin A$. Every $n \in A$ reaches by a possibly empty sequence of even steps to an odd member $m \geq 3$ of A; for odd n, $n \in A \iff \lfloor n^{3/2} \rfloor \in A$; and $A \neq \emptyset$ iff A contains the image $\lfloor m^{3/2} \rfloor$ of some odd $m \geq 3$. In particular every positive integer reaches 1 iff no odd image fails to. (Theorem 6.1; Lean.)

Theorem 3 (the conjecture as an almost-all statement). The following are equivalent: (i) every positive integer reaches 1; (ii) for some $0 < \lambda < \lambda^{**}$, the starts $n \leq x$ whose orbit never enters $[1, N_0]$ have logarithmic count $o((\log x)^{\lambda})$; (iii) for some $e > 1 - \lambda^{**} \approx 0.5074$ and all large y, $\#\{n \text{ odd} \in (y, 2y] : \ n \notin R\} \leq y(\log y)^{-e}$. (Corollary 7.1, Theorems 7.2, 7.3.)

Theorem 4 (the frontier reduction). Let $L(y) = \log_2(\log 2y/\log N_0)$, $d(y) = \lceil CL(y) \rceil$, $p_C = (1 - 1/C)/\log_2 3$, $e(C) = C\,D(p_C \| \frac{1}{2})/\ln 2$, and $e_q^{\mathrm{Az}}(C) = 2(C(1 - q\log_2 3) - 1)^2/(C(\log_2 3)^2 \ln 2)$. Assume $C \geq 5$. The following hypotheses give the indicated, case-dependent bounds. Each implies (iii) of Theorem 3 whenever its displayed rate exceeds $1 - \lambda^{**}$:

(a) *cylinder form* $\mathrm{H}(C, A)$: no O-rooted, $L(y)$-bad itinerary cylinder of depth $d(y)$ exceeds its fair share $2^{-(d-1)}y/2$ among odd starts by more than $y(\log y)^{-A}$, $A > C + e(C)$ (Theorem 8.3), giving $e(C) - \varepsilon$ and hence the conjecture for $C \geq 19$ using Theorem 1 (or $C \geq 18$ under Appendix C);

(b) *one-sided form* $\mathrm{H}_q(C, A)$, with $0 < q < \log 2/\log 3$ and $C > 1/(1 - q\log_2 3)$: every $L(y)$-bad cylinder of depth $1 \leq t < d(y)$ sends at most a fraction q of its members, plus $y(\log y)^{-A}$, to an odd next state (Theorem 9.1; $A > C + e_q^{\mathrm{Az}}(C)$), giving $e_q^{\mathrm{Az}}(C) - \varepsilon$; at $q = 0.55$, $C \geq 41$ crosses Theorem 1's threshold;

(c) *pressure form* $\mathrm{P}_\theta(C)$: $\frac{1}{N}\sum_{n \text{ odd} \in (y, 2y],\ \tau(n) > d} e^{\theta o_d(n)} \leq (\frac{1}{2}(1 + e^{\theta}))^d e^{o(d)}$ at $\theta = \log(p_C/(1 - p_C))$, with $N = y/2$, where τ is the entrance time into $[1, N_0]$ and o_d the odd count of the first d letters; this gives $e(C) - \varepsilon$. Its biased *no-momentum form*, for $0 < q < p_C < 1$, $\sum_{t<d}(s_\theta(t) - q)^+ = o(d)$ gives $CD(p_C \| q)/\ln 2 - \varepsilon$ only at the optimizing tilt $\theta = \log(p_C(1 - q)/(q(1 - p_C)))$ (Theorem 9.2, Proposition 9.3).

None of these hypotheses is proved here. A bounded loss at each of $o(\log\log y)$ exceptional depths can be absorbed in the moment criterion. The tower comparison and finite-depth model in Section 9.3 describe the scope of this argument; neither is an arithmetic necessity theorem.

Theorem 5 (first-letter decomposition). A two-way closed set has the disjoint first-letter decomposition of Section 6.2. In the normalization used there, its logarithmic density satisfies (6.1), with the stated boundary error. For the failure set F, the free term ψ_F is a known normalization factor times the decreasing limit of the live proportion among OO-type starts (Proposition 10.1). Each sufficient estimate of Theorem 4 bounds this term at its stated logarithmic rate. The upper recursion obtained by bounding the fiber weights gives decay estimates with a critical exponent; that upper inequality alone does not force the failure set to be empty (Corollary 10.2).

Proposition (depth-uniformity budget). Suppose the error bound available for each cylinder is $K2^{\kappa d}y^{1-\delta_0 2^{-cd}}$, with $K, \delta_0, c > 0$ and $\kappa \geq 0$ independent of the depth. The resulting union-bound error is negligible at depth $C\log_2\log y + O(1)$ precisely when $cC < 1$ (Proposition 10.3). This is a criterion for the available error majorant, not a lower bound on the actual discrepancy.

Under a localization of the triple parity discrepancy of nested floor powers to sub-dyadic intervals — stated as an explicit hypothesis in Appendix C; it is Theorem 4.12 of the working draft [12], and its status is described there — the exponent of Theorem 1 improves to $\lambda^{***} \approx 0.5392$, the rate threshold of Theorem 3 to 0.4608, and the least depth constant of Theorem 4 to $C \geq 18$. Nothing in Sections 2–12 depends on Appendix C.

1.2 The three fates as currently constrained

The three fate classes are the three Moirai: Atropos cuts the thread (the class R), Lachesis measures an allotted length (a cycle basin), Clotho spins without end (divergence). What is known about each, all from outside this paper:

- *Atropos.* Every $n \leq N_0 = 3.5 \cdot 10^8$ reaches 1 (independently certified computation, [11, Section 5]); the Lean-verified floor is $N_0 = 260$. Paper B [12] proves that 7/8 of all starts descend below themselves within five steps, with a power saving on dyadic blocks.
- *Lachesis.* A nontrivial cycle has minimum above $3.5 \cdot 10^8$ and period at least 780239 [11]; it has at least four even steps; if $3^o/2^L = 1 + \Lambda$, its odd share is $\log 2/\log 3 + \log(1 + \Lambda)/(L \log 3)$, strictly above but, under the finance bound, very near the critical zero-drift share.
- *Clotho.* A divergent orbit has unbounded exponent walk; its record jumps are quantized to the $\log_2 3$ lattice [11]. Nothing excludes it.

Theorem 1 adds one sentence to each: whichever of the three acts at all acts on a set of logarithmic count $\gg (\log x)^\lambda$ for every $0 < \lambda < \lambda^{**}$.

1.3 Related work

Collatz parity vectors. Terras [4] and Everett [5] showed that the first k parities of the Collatz orbit of n are determined by $n \bmod 2^k$ and equidistributed exactly, and deduced that almost all starts have finite stopping time (descend below themselves). The analogous exact statement fails for the Juggler map: the parities of $\lfloor n^{3/2} \rfloor$, $\lfloor \lfloor n^{3/2} \rfloor^{3/2} \rfloor$, ... are Diophantine, not algebraic, and their equidistribution is proved only to depth four (Paper B, [12]).

Tao's theorem. Tao [6] proved that for any $f(N) \to \infty$, almost all Collatz orbits (in logarithmic density) attain a value below $f(N)$. The bounded-target version is not available, and the renewal machinery through the Syracuse random variables is the heart of the proof. Section 7.1 explains, as a structural analogy and not as a transfer of methods, which parts of that picture change for the Juggler map.

Preimage trees. Krasikov and Lagarias [7] obtain an $x^{0.84}$ lower bound for Collatz preimage counts, for targets not divisible by 3 and sufficiently large x. Such a lower bound alone does not give a divergent logarithmic count: a set whose counting function is $O(x^{0.84})$ has a convergent reciprocal sum. It therefore cannot by itself contradict a failure estimate of size $y(\log y)^{-e}$. For Juggler, Theorem 1 gives a divergent logarithmic lower bound for every nonempty preimage set.

Floor-power maps and exponential sums. The parity of $\lfloor n^{3/2} \rfloor$ along an arithmetic progression is a question about $\{n^{3/2}/2\}$; Section 4 uses only an elementary sweep argument and, on even blocks, the second-derivative test with Vaaler's approximation [10], as for Piatetski-Shapiro sequences. Deeper nesting is Paper B's subject.

Companion papers. Paper A [11] proves period lower bounds for a hypothetical cycle from a certified descent floor (cycle financing, walk-charge envelope); Paper B [12] proves parity equidistribution of nested floor powers to depth four with power savings. Neither is reproved here; both enter as citations, and one statement of the type Paper B proves enters Appendix C as an explicit hypothesis.

1.4 Verification

Statements carry one of four roles. *Lean*: the exact combinatorial layer is formalized in Lean 4; the root `formal/Problems/JugglerFatePaper.lean` imports exactly the twelve modules this paper cites and builds with `lake build Problems.JugglerFatePaper`, without `sorry` and without `native_decide`; `formal/AxiomCheckPaperC.lean` prints the axiom dependencies of every cited name and `AxiomCheckPaperC.expected` records them — Mathlib's `propext`, `Classical.choice`, `Quot.sound` and nothing else (names in Appendix A). *Human proof*: the analytic and probabilistic counting. *Verified computation*: exact integer computations (the descent floor is Paper A's; the closure of $[1, 260]$ under the two productions up to 10^9 is computed here). *Observation*: numerical experiments that check hypotheses and constants; they prove nothing and are labelled wherever they appear.

Result	Status
Fate classes closed; trichotomy; exclusion (Lemma 2.1)	Lean
Even block is an interval (Lemma 3.1)	Lean
OE fiber is an interval; cell identity (Lemma 3.2)	Lean
Odd generation (Theorem 6.1)	Lean
Envelope descent into the floor (Lemma 8.1)	Lean, on Paper A's power envelope
Sweep lemma (Lemma 4.1), both half-cell conventions	Lean
Monotone pairing (Lemma 4.1')	Lean
Fiber parity, thin fibers (Lemmas 4.2–4.3)	human proof
Block average (Proposition 4.4), asymptotic	human proof
Share law and its three corollaries (Lemma 4.5, Corollary 4.6)	human proof
Cube fibers full or alternating (Lemma 4.7)	Lean
Recursion lemma (Lemma 5.1)	Lean
Seed (Lemma 5.2)	Lean
Contagion (Theorem 5.3)	human proof
Least failure is OO-type; first-letter trichotomy (Proposition 6.3(i), Section 6.2)	Lean
First-letter identity (6.1)	human proof (exact combinatorics)
Tao-type rate implies the conjecture (Theorem 7.2), with the contagion bound of Theorem 5.3 as a hypothesis	Lean
Almost-all equivalence (Theorem 7.3)	human proof
Chernoff count of bad words (Lemma 8.2, exact form)	Lean
Theorem 8.3, explicit form $y\Lambda^{-e(C)} + 2\Lambda^C y(\log y)^{-A}$ at every y	Lean; the ε-absorption into the displayed form is human
Pressure form (Theorem 9.2, exact: a pressure bound $Na_\theta^d E$ gives at most $Ne^{-dD(p_C\|1/2)}E$ live starts)	Lean, on the live weight
Pressure telescoping (Proposition 9.3)	Lean, on the word-weight framework
Azuma and exponential-moment arguments (Sections 8–10, except Lemma 8.2, Theorem 8.3, Theorem 9.2 and Proposition 9.3 in their exact forms)	human proof
Localized triple discrepancy (Appendix C)	hypothesis, conditional
Numerical experiments (Section 11)	observation

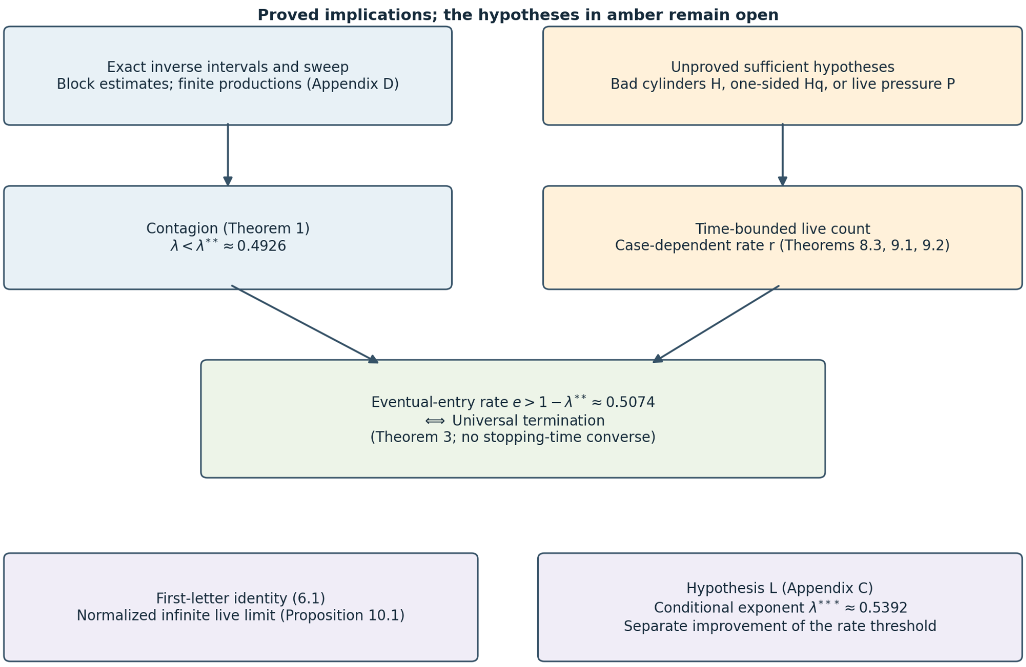

Figure 1: Logical dependencies. The arithmetic lemmas feed the contagion theorem. Each unproved parity hypothesis is a sufficient route to a time-bounded live count, which implies eventual termination only at the stated rate. No converse time bound is asserted.

2. Fate classes and closure

$\mathbb{N} = \{1, 2, \ldots\}$. For $A \subseteq \mathbb{N}$ and $0 \le u < v$ write

$$\ell_A(u, v] = \sum_{\substack{n \in A \\ u < n \le v}} \frac{1}{n}, \qquad g_A(t) = \ell_A\big(e^{t/2}, e^t\big].$$

For $A = \mathbb{N}$, $g(t) = t/2 + O(e^{-t/2})$.

Definition. A is *backward-closed* if $J(n) \in A$ implies $n \in A$, and *forward-closed* if $n \in A$ implies $J(n) \in A$.

Lemma 2.1 (fate classes). Let $R = \{n : \exists k,\ J^k(n) = 1\}$, $F = \mathbb{N} \setminus R$, $B(m) = \{n : \exists k,\ J^k(n) = m\}$, and $D = \{n : \forall B\ \exists k,\ J^k(n) > B\}$. Each of $R, F, B(m), D$ is backward-closed; R, F, D are also forward-closed. Every $n \ge 1$ lies in R, or in $B(m)$ for some $m \ge 2$ on a cycle, or in D, and these possibilities exclude one another.

Proof. Backward closure is the definition of each set (for D: $J^k(J(n)) = J^{k+1}(n)$). Forward closure of R: if $J^k(n) = 1$ then $J^{k-1}(J(n)) = 1$ for $k \ge 1$, and $J(1) = 1$. Forward closure of D: given B, pick k with $J^k(n) > \max(B, n)$; then $k \ge 1$ and $J^{k-1}(J(n)) > B$. Trichotomy: a bounded orbit repeats, a repeat is a cycle, a cycle through 1 is the fate R. Exclusion: an orbit that reaches 1 or enters a cycle is bounded; an orbit that reaches 1 and passes through a cycle state $m \ge 2$ of period L would force $m = J^{qL}(m) = 1$. Lean: `reachesOne_backwardClosed`, `not_reachesOne_backwardClosed`, `ancestor_backwardClosed`, `escapes_backwardClosed`, `reachesOne_floorPower`, `escapes_floorPower`, `fate_trichotomy`, `reachesOne_not_escapes`, `cycle_basin_not_escapes`, `reachesOne_not_cycle_basin`. ☐

Throughout Sections 3–5, A is a nonempty backward-closed set. Every statement applies to each fate class realized by at least one start.

3. The two productions

Lemma 3.1 (even block). Let $m \in A$. Every even n with $m^2 \le n < (m+1)^2$ lies in A. Writing $E(m)$ for this set, $|E(m)| \ge m$ and

$$\frac{1}{m}\Big(1-\frac{2}{m}\Big) \;\le\; \frac{m}{(m+1)^2} \;\le\; \sum_{n\in E(m)} \frac{1}{n} \;\le\; \frac{m+1}{m^2}.$$

Proof. $J(n) = \lfloor\sqrt{n}\rfloor = m$ exactly on that interval, so $n \in A$. The interval has $2m+1$ integers, at least m and at most $m+1$ of them even, each between m^2 and $(m+1)^2$. Lean: `even_block_mem`, `even_block_card`. □

Lemma 3.2 (*OE* fiber). Let $m \in A$ and

$$\Phi(m) = \{n \text{ odd}: \; m^4 \le n^3 < (m+1)^4\} = \{n \text{ odd}: \; m^{4/3} \le n < (m+1)^{4/3}\}.$$

If $n \in \Phi(m)$ and $\lfloor n^{3/2}\rfloor$ is even, then $J(J(n)) = m$, hence $n \in A$. Moreover $H_m := |\Phi(m)| \in [\frac{2}{3}m^{1/3} - 1,\; \frac{2}{3}(m+1)^{1/3} + 1]$, and the fibers of distinct m are disjoint.

Proof. $J(n) = \lfloor\sqrt{n^3}\rfloor =: k$; if k is even then $J(k) = \lfloor\sqrt{k}\rfloor$, and $\lfloor\sqrt{\lfloor\sqrt{N}\rfloor}\rfloor = m \iff m^4 \le N < (m+1)^4$: this is the exact form of the transparent nesting $\lfloor\sqrt{\lfloor n^{3/2}\rfloor}\rfloor = \lfloor n^{3/4}\rfloor$. The interval $[m^{4/3}, (m+1)^{4/3})$ has length between $\frac{4}{3}m^{1/3}$ and $\frac{4}{3}(m+1)^{1/3}$ by the mean value theorem, and a half-open interval of length ℓ contains between $\ell/2 - 1$ and $\ell/2 + 1$ odd integers. Disjointness is the cell identity. Lean: `sqrt_sqrt_eq_iff`, `oe_fiber_mem`, `floorPower_oe_fiber`, `oe_fiber_disjoint`. □

Write $G_m = \#\{n \in \Phi(m) : \lfloor n^{3/2}\rfloor \text{ even}\}$. The whole argument rests on lower bounds for G_m: an elementary bound on every fiber outside a thin exceptional set (Section 4.1), and a classical average over an even block of m's (Section 4.2).

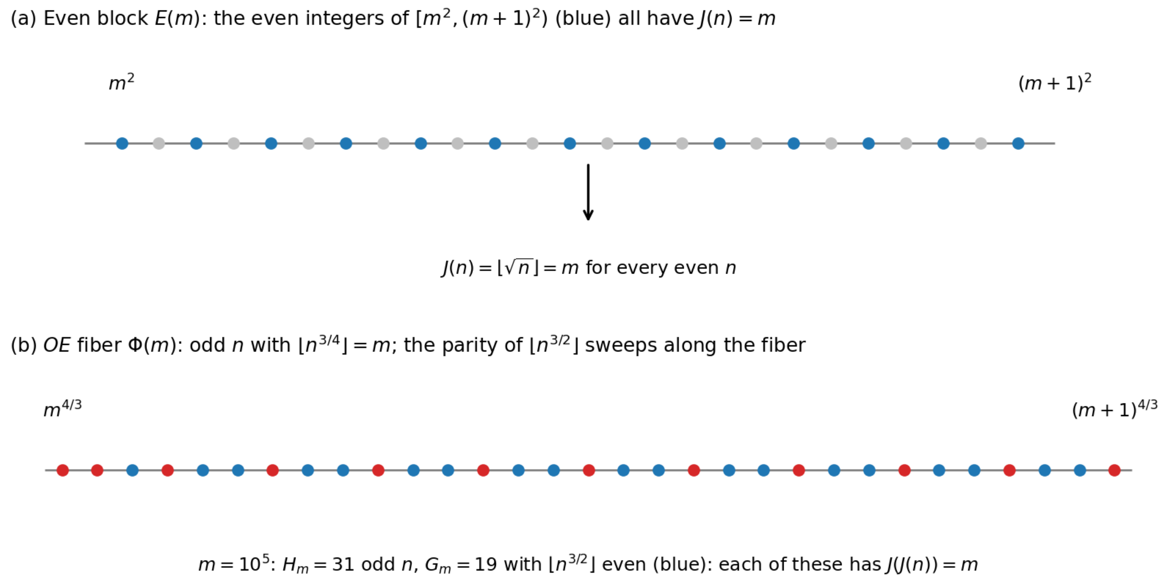

Figure 2: The two productions. (a) The even integers of $[m^2, (m+1)^2)$ all map to m. (b) The *OE* fiber $\Phi(m)$ for $m = 10^5$: the odd n with $\lfloor n^{3/4}\rfloor = m$, coloured by the parity of $\lfloor n^{3/2}\rfloor$; those with even image (blue) have $J(J(n)) = m$.

4. Parity on a fiber

4.1 The sweep lemma

On $\Phi(m)$ the quantity $x = n^{3/2}/2$ has $\lfloor n^{3/2} \rfloor$ even iff $\{x\} < \frac{1}{2}$. Consecutive odd n advance x by $\approx \frac{3}{2}n^{1/2} \approx \frac{3}{2}m^{2/3}$, and over the whole fiber this step drifts by less than one unit of $1/H_m$. So modulo 1 the fiber is an arithmetic progression with a nearly constant step α, of length H_m. Such a progression cannot avoid a half-circle unless its step is within $O(1/H_m)$ of 0, or its two-step is ($\alpha \approx \frac{1}{2}$ with the wrong phase).

Lemma 4.1 (sweep). Let $x_1 < x_2 < \cdots < x_H$ be real numbers with $x_{j+1} - x_j \in [a, b]$ for all j, where $0 < a \le b \le \frac{1}{2}$, $b \le \frac{21}{20}a$ and $(H-1)a \ge 12$. Then

$$\#\{j : \{x_j\} < \tfrac{1}{2}\} \ \ge \ \tfrac{H}{7} \qquad \text{and} \qquad \#\{j : \{x_j\} \ge \tfrac{1}{2}\} \ \ge \ \tfrac{H}{7}.$$

The same holds with the cells $(k/2, (k+1)/2]$ in place of $[k/2, (k+1)/2)$, i.e. with the representative in $(0, 1]$ in place of the fractional part.

Proof. Cut $\mathbb{R}$ into cells $C_k = [k/2, (k+1)/2)$, $k \in \mathbb{Z}$; even k are the *good* cells ($\{x\} < \frac{1}{2}$), odd k the *bad* cells. Call C_k *traversed* if $x_1 \le k/2$ and $(k+1)/2 \le x_H$.

(i) *A traversed cell contains at least $g := \lfloor 1/(2b) \rfloor \ge 1$ and at most $G := \lfloor 1/(2a) \rfloor + 1$ of the points.* Let j_0 be the least index with $x_{j_0} \ge k/2$. Then $x_{j_0} < k/2 + b$ (either $j_0 = 1$ and $x_1 = k/2$, or $x_{j_0 - 1} < k/2$). Since every step is at most b, $x_{j_0+i} < k/2 + (i+1)b \le (k+1)/2$ for all $i \le 1/(2b) - 1$, and these indices exist because $(k+1)/2 \le x_H$. That gives at least $\lfloor 1/(2b) \rfloor$ points in C_k. Since every step is at least a, the points in C_k are x_{j_0+i} with $k/2 + ia \le x_{j_0+i} < (k+1)/2$, so $i < 1/(2a)$ and there are at most $\lfloor 1/(2a) \rfloor + 1$ of them. The same count holds for any cell as an upper bound.

(ii) *Counting cells.* The traversed cells are consecutive, and their number T is at least $2(x_H - x_1) - 2 \ge 2(H-1)a - 2 \ge 22$. Good and bad traversed cells alternate, so their numbers T_g, T_b differ by at most one and $T_g, T_b \ge 10$. Every point outside the traversed cells lies in the cell of x_1 or the cell of x_H.

(iii) *The ratio.* Hence $\text{Good} \ge gT_g$ and $H \le G(T_g + T_b + 2) \le G(2T_g + 3)$, so

$$\frac{\text{Good}}{H} \ \ge \ \frac{g}{G} \cdot \frac{T_g}{2T_g + 3} \ \ge \ \frac{g}{G} \cdot \frac{10}{23}.$$

Put $X = 1/(2a) \ge 1$, so $G = \lfloor X \rfloor + 1$ and, from $b \le \frac{21}{20}a$, $g \ge \lfloor \frac{20}{21}X \rfloor$. If $X < 2$: $g \ge 1, G \le 2$. If $2 \le X < 3$: $g \ge 1, G = 3$. If $3 \le X < 4$: $g \ge 2, G = 4$. If $4 \le X < 5$: $g \ge 3, G = 5$. If $X \ge 5$: $g/G \ge (\frac{20}{21}X - 1)/(X+1) \ge 0.62$. In all cases $g/G \ge \frac{1}{3}$, so $\text{Good}/H \ge \frac{10}{69} > \frac{1}{7}$. The bad count is symmetric. For left-open cells the same proof applies verbatim (the first point in a cell $(c, c + \frac{1}{2}]$ after a point $\le c$ is $\le c + b \le c + \frac{1}{2}$). $\square$

Lean: `sweep_fract_lt_half`, `sweep_fract_ge_half` (closed cells) and `sweep_rep_le_half`, `sweep_rep_gt_half` (left-open cells, with the representative $x - \lceil x \rceil + 1 \in (0, 1]$), in `formal/Problems/Juggler/FateSweep.lean`. The formal count uses the strictly interior cells (at least $(T-3)/2$ of one colour, ratio 11/25) in place of the traversed cells (ratio 10/23), and obtains the left-open case from the closed one by $x_j \mapsto -x_{H+1-j}$.

The product $g/G \cdot 10/23$ cannot reach $\frac{1}{3}$. An adversarial sequence with steps in $[a, b]$ can lock a $3 + 1$ split near $a = \frac{1}{4}$. The OE fiber is monotone.

Lemma 4.1' (monotone pairing). Keep the hypotheses of Lemma 4.1 and assume the consecutive differences are monotone (nondecreasing or nonincreasing). Then

$$\#\{j : \{x_j\} < \tfrac{1}{2}\} \ \ge \ \tfrac{H}{3} - 2 \qquad \text{and} \qquad \#\{j : \{x_j\} \ge \tfrac{1}{2}\} \ \ge \ \tfrac{H}{3} - 2.$$

The same holds with the cells $(k/2,(k+1)/2]$ in place of $[k/2,(k+1)/2)$.

Proof. We prove the nondecreasing case for both half-cell conventions. Reversing the order and reflecting, $z_j = -x_{H+1-j}$, then gives the nonincreasing case; it interchanges the half-cell conventions and preserves the two color counts up to relabeling. Assume the steps are nondecreasing. Cut $\mathbb{R}$ into the cells $C_k = [k/2,(k+1)/2)$ of Lemma 4.1. Since every step is at most $b \le \frac{1}{2}$, the cells of $x_1,\dots,x_H$ are T consecutive integers, each visited; write $\rho_1,\dots,\rho_T \ge 1$ for their occupancies in order, so $\sum_i \rho_i = H$, and consecutive cells have opposite colours. Call the cells $2,\dots,T-1$ *interior.* Put $X = 1/(2a)$, $G = \lfloor X \rfloor + 1$ and $g = \lfloor 1/(2b) \rfloor \ge 1$. As in Lemma 4.1(i), every cell has $\rho_i \le G$ and every interior cell has $\rho_i \ge g$; and $(H-1)a \ge 12$ gives $H \ge 24X + 1 \ge 24G - 23$.

(a) *A later cell is at most one fuller than an earlier interior cell.* Let $2 \le i < j \le T$. The $\rho_i + 1$ steps from the last point before the cell of index i to the first point after it span more than $\frac{1}{2}$, so one of them, γ, exceeds $1/(2(\rho_i+1))$. Every step inside the cell of index j comes later, hence is at least γ, and the $\rho_j - 1$ steps inside that cell span less than $\frac{1}{2}$. So $(\rho_j - 1)\gamma < \frac{1}{2}$, i.e. $\rho_j \le \rho_i + 1$. In particular, once an interior cell of occupancy v has occurred, every later cell has occupancy at most $v+1$.

(b) *Pairs and surplus.* Pair the interior cells consecutively, $(2,3),(4,5),\dots$; at most three cells are unpaired (the first, the last, and possibly one interior cell), each holding at most G points. Each pair has one cell of each colour, so each colour receives at least $\sum_{\text{pairs}} \min(\rho,\rho')$, and

$$\sum_{\text{pairs}} \min(\rho,\rho') \;\ge\; \tfrac{1}{3}(H-3G) + S, \qquad S := \sum_{\text{pairs}} \Big(\min(\rho,\rho') - \tfrac{\rho+\rho'}{3}\Big).$$

It therefore suffices to show $S \ge G-2$. Both entries of every pair lie in $[g,G]$, so every pair has $\min/\mathrm{sum} \ge g/(g+G) =: r$, and a pair contributes at least $(r - \frac{1}{3})(\rho+\rho')$ to S.

(c) $G \ge 7$. Then $X \ge 6$ and $g \ge \lfloor \frac{20}{21} X \rfloor$ give $7g \ge 5G$ ($g \ge 5$ at $G = 7$, $g \ge 6$ at $G = 8$, and $7g > \frac{20G-41}{3} \ge 5G$ for $G \ge 9$), so $r \ge \frac{5}{12}$ and $S \ge \frac{1}{12}(H-3G) \ge \frac{21G-23}{12} \ge G-2$.

(d) $3 \le G \le 6$ *and* $g \ge 2$. The possible (G,g) are $(3,2),(4,2),(4,3),(5,3),(5,4),(6,4),(6,5)$. Except for $(4,2)$, $r \ge \frac{3}{8}$, so $S \ge \frac{1}{24}(H-3G) \ge \frac{21G-23}{24} \ge G-2$ because $3G \le 25$. For $(G,g) = (4,2)$ the interior occupancies lie in $\{2,3,4\}$; by (a), after the first interior cell of occupancy 2 every cell has occupancy at most 3. So the pairs with both entries in $\{3,4\}$ (ratio $\ge \frac{3}{7}$) precede the pairs with both entries in $\{2,3\}$ (ratio $\ge \frac{2}{5}$), and at most one pair straddles, with ratio $\ge \frac{1}{3}$ and at most 6 points. Hence $S \ge \frac{1}{15}(H-3G-6) \ge \frac{24G-41}{15} > 2 = G-2$.

(e) $G = 2$. Interior occupancies lie in $\{1,2\}$, every pair has ratio $\ge \frac{1}{3}$, and $S \ge 0 = G-2$.

(f) $G = 3$ *and* $g = 1$, i.e. $2 \le X < 3$ and $b > \frac{1}{4}$; then $a \le \frac{1}{4}$ and $b \le \frac{21}{80}$, so $3b < 1$ and $7b < 2$. Interior occupancies lie in $\{1,2,3\}$, and by (a) every cell after the first interior cell of occupancy 1 has occupancy at most 2. The pairs are therefore: pairs with both entries in $\{2,3\}$, of ratio $\ge \frac{2}{5}$; at most one straddling pair $(\rho,1)$ with $\rho \in \{2,3\}$, which contributes at least $-\frac{1}{3}$ to S; and pairs with both entries in $\{1,2\}$. In the last group, $(1,1)$ does not occur: two adjacent interior cells with one point each would have three consecutive steps spanning more than $1 > 3b$. Nor do two consecutive pairs with six points in all: four consecutive interior cells with six points would have seven consecutive steps spanning more than $2 > 7b$. So among any two consecutive pairs of the last group at least one is $(2,2)$, and the two together hold at least 7 points of which the scarcer colour gets at least 3: their contribution to S is at least $3 - \frac{7}{3} = \frac{2}{3}$, and they hold at most 8 points, so the last group contributes at least $\frac{1}{12}N_2 - \frac{2}{3}$ to S, where N_2 is its number of points (a leftover single pair contributes at least 0). With N_1 the number of points in the first group, $N_1 + N_2 \geq H - 3G - 4 = H - 13$, and

$$S \;\geq\; -\tfrac{1}{3} + \tfrac{1}{15}N_1 + \tfrac{1}{12}N_2 - \tfrac{2}{3} \;\geq\; \tfrac{1}{15}(H-13) - 1 \;\geq\; 1 = G - 2,$$

since $H \geq 24X + 1 \geq 49$.

In all cases $S \geq G - 2$, so the scarcer colour has at least $\frac{1}{3}(H - 3G) + G - 2 = \frac{H}{3} - 2$ points. For left-open cells the same argument applies with the cells $(k/2, (k+1)/2]$, or apply the closed case to $-x_{H+1-j}$ as in Lemma 4.1. ☐

Lean: `sweep_monotone_fract_lt_half`, `sweep_monotone_fract_ge_half` (closed cells) and `sweep_monotone_rep_le_half`, `sweep_monotone_rep_gt_half` (left-open cells), instances of `Sweep.sweep_monotone_cell` and `sweep_monotone_ceil`, in `formal/Problems/Juggler/FateSweepMonotone.lean`.

Remark (erratum, 8 September 2026). An earlier version of this proof asserted that every pair satisfies min $\geq (\rho+\rho')/3$, on the grounds that the step scale $1/(2\delta)$ changes by at most $X/21$ "spread over the cells" and that with $X < 2.1$ the pairs are $(1,1)$, $(1,2)$ or $(2,2)$. Monotone steps need not change gradually: with $a = \frac{10}{41}$, $b = \frac{21}{82}$ the sequence $-2a, -a, 0, a, 2a, 2a+b, 2a+2b, \ldots$ has nondecreasing steps in $[a, b]$, and its cells 2 and 3 — an interior pair in the pairing of (b) — have occupancies $(3, 1)$. That version also paired the two partial end cells as if they were interior. The statement is unchanged; the proof above replaces it, with (a) in place of the spreading claim, the end cells left unpaired, and the surplus S paying for them.

Lemma 4.2 (fiber parity). For $m \geq 10^6$ put $\alpha_m = \{\frac{3}{2}m^{2/3}\}$ and call m *good* if

$$\|\alpha_m\| \;\geq\; 22\,m^{-1/3} \qquad \text{and} \qquad \|\alpha_m - \tfrac{1}{2}\| \;\geq\; 2\,m^{-1/3},$$

where $\|\cdot\|$ is the distance to the nearest integer. If m is good then $G_m \geq \frac{1}{3}H_m - 2$ and $H_m - G_m \geq \frac{1}{3}H_m - 2$.

Proof. Let $n_1 < \cdots < n_H$ be the odd integers of $\Phi(m)$, $n_{j+1} = n_j + 2$, and $x_j = n_j^{3/2}/2$. Then $\lfloor n_j^{3/2} \rfloor$ is even iff $\{x_j\} < \frac{1}{2}$. The steps are

$$\delta_j = x_{j+1} - x_j = \int_{n_j}^{n_j+2} \tfrac{3}{4}t^{1/2}\,dt \in \big[\tfrac{3}{2}n_j^{1/2},\ \tfrac{3}{2}(n_j+2)^{1/2}\big] \subseteq [A_m, B_m],$$

with $A_m = \frac{3}{2}m^{2/3}$ and $B_m = \frac{3}{2}\big((m+1)^{4/3} + 2\big)^{1/2}$. Using $\sqrt{u} - \sqrt{v} \leq (u-v)/(2\sqrt{v})$ and $(m+1)^{4/3} - m^{4/3} \leq \frac{4}{3}(m+1)^{1/3}$,

$$\eta_m := B_m - A_m \leq \frac{(m+1)^{1/3} + \frac{3}{2}}{m^{2/3}} \leq m^{-1/3}\Big(1 + \frac{1}{3m} + \frac{3}{2}m^{-1/3}\Big) \leq 1.02\,m^{-1/3}.$$

Also $H_m - 1 \geq \frac{2}{3}m^{1/3} - 2 \geq 0.646\,m^{1/3}$.

Case 1: $\alpha_m \leq \frac{1}{2} - 2m^{-1/3}$. Put $N = \lfloor A_m \rfloor$ and $y_j = x_j - jN$; then $\{y_j\} = \{x_j\}$ and the steps of y lie in $[a, b] = [\alpha_m, \alpha_m + \eta_m] \subseteq [22m^{-1/3}, \frac{1}{2}]$. Now $b/a \leq 1 + 1.02/22 < \frac{21}{20}$ and $(H-1)a \geq 0.646 \cdot 22 > 12$. The steps of y are nondecreasing. Lemma 4.1' applies.

Case 2: $\alpha_m \geq \frac{1}{2} + 2m^{-1/3}$. Put $z_j = j(N+1) - x_j$, increasing with steps $(N+1) - \delta_j \in [1 - \alpha_m - \eta_m,\ 1 - \alpha_m] \subseteq [20.98\,m^{-1/3},\ \frac{1}{2}]$; again $b/a \leq 1 + 1.02/20.98 < \frac{21}{20}$ and $(H-1)a \geq 0.646 \cdot 20.98 > 12$. Write $\langle z \rangle \in (0,1]$ for the representative of z modulo 1 in $(0,1]$. Then $\langle z_j \rangle = 1 - \{x_j\}$ for every j, so $\{x_j\} < \frac{1}{2} \iff \langle z_j \rangle \in (\frac{1}{2}, 1]$. The steps of z are monotone. Lemma 4.1' in its left-open form gives both counts.

The middle range $|\alpha_m - \frac{1}{2}| < 2m^{-1/3}$ and the range $\|\alpha_m\| < 22m^{-1/3}$ are excluded by goodness. ☐

Lemma 4.3 (bad fibers are thin). For $u \geq 10^6$,

$$\#\{m \in (u, 2u] : \ m \text{ bad}\} \leq 63\, u^{2/3}, \qquad \sum_{\substack{m>U \\ m \text{ bad}}} \frac{1}{m} \leq 306\, U^{-1/3} \quad (U \geq 10^6).$$

Proof. $\varphi(m) = \frac{3}{2}m^{2/3}$ increases with $\varphi(m+1) - \varphi(m) \in [(m+1)^{-1/3}, m^{-1/3}] \subseteq [(3u)^{-1/3}, u^{-1/3}]$ on $(u, 2u]$. Badness means $\{\varphi(m)\}$ lies in one of two arcs of the circle, of total length at most $48\, u^{-1/3}$. As m runs over $(u, 2u]$, φ increases by $\frac{3}{2}u^{2/3}(2^{2/3} - 1) < 0.882\, u^{2/3}$, so it passes each arc at most $0.882u^{2/3} + 2$ times, and each pass through an arc of length w takes at most $w(3u)^{1/3} + 1$ values of m. Hence the count is at most $(0.882u^{2/3} + 2)(48 \cdot 3^{1/3} + 2) \leq 63\, u^{2/3}$ for $u \geq 10^6$. Summing $63\,(2^i U)^{-1/3}$ over $i \geq 0$ gives $63U^{-1/3}/(1 - 2^{-1/3}) \leq 306\, U^{-1/3}$. $\square$

4.2 The block average

The adversarial sweep $G_m \geq H_m/7$ is not the fiber bound: Lemma 4.1' gives the monotone floor $H_m/3 - 2$. In the numerical experiments of Section 11 the mean of G_m/H_m is $\frac{1}{2}$ and the minimum on good fibers is near $\frac{1}{3}$. When the members of A come in even blocks — as the E-produced part of A always does — the fibers can be averaged over the block, and the average is asymptotically $\frac{1}{2}$ by a classical exponential-sum estimate.

Proposition 4.4 (block average). For $m' \geq 2$, put $I(m') = [m'^{8/3}, (m'+1)^{8/3})$ and

$$U(m') = \{n \text{ odd in } I(m') : \lfloor n^{3/4} \rfloor \text{ even}, \quad \lfloor n^{3/2} \rfloor \text{ even}\}.$$

This is the disjoint union of the even-image parts of $\Phi(m)$ over $m \in E(m')$. There is an absolute constant C_B such that

$$\big| |U(m')| - \tfrac{1}{4}\#\{n \text{ odd in } I(m')\} \big| \leq C_B m'^{11/9} \log(m'+1). \tag{4.1}$$

In particular

$$|U(m')| = \tfrac{1}{3}m'^{5/3}(1 + O(m'^{-4/9}\log(m'+1))).$$

No numerical value of C_B is asserted.

Proof. Write $M = \#\{n \text{ odd in } I(m')\} \asymp m'^{5/3}$ and $\psi(z) = (-1)^{\lfloor z \rfloor}$. Expanding the two indicators gives four sums: M, the sums of $\psi(n^{3/4})$ and $\psi(n^{3/2})$, and their product.

The slow sum is $O(m')$: pair consecutive fibers of $\lfloor n^{3/4} \rfloor$, whose odd-point counts differ by at most 3, and bound the remaining end fiber by $O(m'^{2/3})$.

We use the interval form of Vaaler's approximation [10]. For $J \geq 1$, the parity wave has a trigonometric approximation $V_J(z/2)$, with zero constant term and coefficients $O(1/|q|)$, and a nonnegative error majorant $\Delta_J(z/2)$. Its degree is J, every coefficient is $O(1/J)$, and $\|\Delta_J\|_\infty = O(1)$. All constants in this paragraph are absolute; the interval endpoints are included in the majorant.

Take $J = \max(1, \lfloor m'^{4/9} \rfloor)$ and reindex odd integers as $n = 2r+1$. For $q \neq 0$, the fast phase $f(r) = q(2r+1)^{3/2}/2$ has $|f''| \asymp |q| m'^{-4/3}$, with constant sign. The second-derivative test [13, Theorem 2.2] gives, uniformly on initial subintervals,

$$\left| \sum_r e(f(r)) \right| \ll m'|q|^{1/2} + m'^{2/3}|q|^{-1/2}, \qquad e(z) = e^{2\pi i z}.$$

The coefficient sum and the majorant sum are therefore bounded by

$$\ll m' J^{1/2} + m'^{2/3} + M/J \ll m'^{11/9}.$$

For the product, expand both waves. A mixed phase is $f = (q_1 n^{3/4} + q_2 n^{3/2})/2$, with $q_2 \neq 0$, and

$$f''(r) = \tfrac{3}{2}q_2 n^{-1/2}\left(1 - \frac{q_1}{4q_2}n^{-3/4}\right).$$

For $|q_1| \le J$, the correction is $O(m'^{-14/9})$; for all sufficiently large m' the same second-derivative estimate applies uniformly. Summing the slow coefficients adds at most $O(\log(J+1))$.

The slow majorant has constant contribution $O(M/J)$. Its nonzero modes have monotone derivative $\frac{3}{4}q_1 n^{-1/4}$, of magnitude $\asymp |q_1|m'^{-2/3}$ and at most $1/2$ for sufficiently large m'. Kusmin–Landau [13, Theorem 2.1] bounds a mode by $O(m'^{2/3}/|q_1|)$, so the majorant contributes $O(M/J + m'^{2/3}\log(J+1)/J)$. Since $|\psi| = 1$, $|V_J| \le 1 + \Delta_J = O(1)$, and the majorants are nonnegative, the product-approximation error is bounded by a constant times the sum of the two majorants. These estimates prove (4.1) for sufficiently large m'; increasing the unspecified absolute constant covers the finitely many smaller integers $m' \ge 2$. □

Revision note. The earlier value $C_0 = 250$ depended on an unverified explicit majorant and small-parameter inequalities that do not hold throughout their printed range. The asymptotic form above is all that the subsequent recursions use.

4.3 The share is a quadratic sweep

Sections 4.1 and 4.2 bound G_m from below, by a sweep on each good fiber and by an exponential sum across an even block. Neither says what G_m/H_m *is*. It has a closed form, and the form explains both the floor of Lemma 4.2 and the extreme fibers that the goodness condition excludes. Nothing in Sections 5–10 depends on this subsection; it is here because it identifies which of the two tools is doing the work, and because it names the hypothesis behind the gap between them.

Throughout, $n_1 = \min \Phi(m)$, $n_j = n_1 + 2(j-1)$, and $x_j = n_j^{3/2}/2$, so that $\lfloor n_j^{3/2} \rfloor$ is even iff $\{x_j\} < \frac{1}{2}$. Write

$$\theta_m = \{x_1\}, \qquad \beta_m = \alpha_m\,(H_m - 1),$$

with $\alpha_m = \{\frac{3}{2}m^{2/3}\}$ as in Lemma 4.2, taken in $(-\frac{1}{2}, \frac{1}{2}]$, and set

$$S(\beta,\theta) = \left|\{s \in [0,1] :\ \{\theta + \beta s + \tfrac{1}{3}s^2\} < \tfrac{1}{2}\}\right|.$$

Lemma 4.5 (share law). As $m \to \infty$,

$$G_m/H_m = S(\beta_m, \theta_m) + O\left(H_m^{-1/2} + \frac{1 + |\beta_m|}{H_m}\right).$$

The estimate is useful when $|\beta_m| = o(H_m)$; it is not a uniform equidistribution assertion for all fibers.

Proof. Expand x_j about n_1. With $u = 2(j-1)/n_1$,

$$x_j = \tfrac{1}{2}n_1^{3/2}(1+u)^{3/2} = x_1 + \tfrac{3}{2}n_1^{1/2}(j-1) + \tfrac{3}{4}n_1^{-1/2}(j-1)^2 + E_j, \qquad |E_j| \le \tfrac{1}{4}(j-1)^3 n_1^{-3/2}.$$

On the fiber $j - 1 \le H_m - 1 \ll m^{1/3}$ and $n_1 \ge m^{4/3}$, so $|E_j| \ll m^{-1}$, below $1/H_m$. Put $s = (j-1)/(H_m - 1) \in [0,1]$. The linear coefficient is $A = \frac{3}{2}n_1^{1/2}$, and since $n_1 = m^{4/3} + O(1)$ we have $A = \frac{3}{2}m^{2/3} + O(m^{-2/3})$, so $A \equiv \alpha_m + O(m^{-2/3}) \pmod 1$ and the accumulated difference over the fiber is $O(m^{-1/3}) = O(1/H_m)$. Hence $A(j-1) \equiv \beta_m s \pmod 1$ up to $O(1/H_m)$. The quadratic coefficient gives $\frac{3}{4}n_1^{-1/2}(H_m - 1)^2 = \frac{1}{3} + O(1/H_m)$ by Lemma 3.2, so

$$x_j \ \equiv\ \theta_m + \beta_m s + \tfrac{1}{3}s^2 \pmod 1,$$

uniformly to $O(1/H_m)$. Put $\epsilon = O(1/H_m)$. Replacing the exact phase by the displayed quadratic can change an indicator only within ϵ of an integer or half-integer. If $|\beta_m| \le 2$, there are $O(1)$ such levels and their inverse images have total length $O(\sqrt{\epsilon})$, since the quadratic has second derivative $2/3$. If $|\beta_m| > 2$, its derivative has magnitude at least $|\beta_m| - 2/3$, and the total inverse-image length is $O(\epsilon)$. Each of these sets, and each half-cell inverse image, has $O(1 + |\beta_m|)$ interval components. Sampling on the uniform grid costs $O((1 + |\beta_m|)/H_m)$. This proves the stated bound, including tangencies to the cell boundaries. □

Corollary 4.6. For every β:

1. $\int_0^1 S(\beta,\theta)\,d\theta = \frac{1}{2}$.
2. $S(\beta,\theta) \in \{0,1\}$ for some θ if and only if $\beta \in [-\frac{5}{6}, \frac{1}{6}]$.
3. $\displaystyle\int \big|\{\theta : S(\beta,\theta) = 0\}\big|\,d\beta = \frac{25}{108}$, and the same for $S = 1$.

Proof. (1) By Fubini the integral is $\int_0^1\int_0^1 \mathbf{1}[\{\theta + \varphi(s)\} < \frac{1}{2}]\,d\theta\,ds$, and the inner integral is $\frac{1}{2}$ for every s, whatever φ.

(2) Let $\varphi(s) = \beta s + \frac{1}{3}s^2$, so $\varphi'(s) = \beta + \frac{2}{3}s$. If $\beta \geq 0$ then φ is increasing with range $\beta + \frac{1}{3}$; if $\beta \leq -\frac{2}{3}$ it is decreasing with range $-\beta - \frac{1}{3}$; and if $-\frac{2}{3} < \beta < 0$ it dips to $\varphi(-\frac{3}{2}\beta) = -\frac{3}{4}\beta^2$ and returns to $\beta + \frac{1}{3}$, so its range is $\max(0, \beta + \frac{1}{3}) + \frac{3}{4}\beta^2$. An extreme value of S needs the image of $[0,1]$ to miss a half-circle, hence range $\leq \frac{1}{2}$. That is $\beta \leq \frac{1}{6}$ in the first case, $\beta \geq -\frac{5}{6}$ in the second, and no constraint in the third, since the range there is at most $\frac{1}{3}$.

(3) The θ-measure is $\max(0, \frac{1}{2} - \text{range})$. Integrating the four pieces of (2),

$$\int_0^{1/6}\big(\tfrac{1}{6}-\beta\big)+\int_{-1/3}^{0}\big(\tfrac{1}{6}-\beta-\tfrac{3}{4}\beta^2\big)+\int_{-2/3}^{-1/3}\big(\tfrac{1}{2}-\tfrac{3}{4}\beta^2\big)+\int_{-5/6}^{-2/3}\big(\tfrac{5}{6}+\beta\big) = \tfrac{1}{72}+\tfrac{11}{108}+\tfrac{11}{108}+\tfrac{1}{72} = \tfrac{25}{108}.$$

The statement for $S = 1$ is the reflection $\theta \mapsto \theta + \frac{1}{2}$. ☐

Part (1) is the reason no aggregate count in this paper sees the extreme fibers: they are compensated at every drift separately, which is what Proposition 4.4 recovers by a different route. Part (2) says the extremes occupy a window of one drift unit, that is $\alpha_m \in [-\frac{5}{4}, \frac{1}{4}]m^{-1/3}$ by $H_m \sim \frac{2}{3}m^{1/3}$.

At the centre of that window the law degenerates to an identity.

Lemma 4.7 (cube fibers). Let $k \geq 2$ and $m = k^3$. Every $n \in \Phi(m)$ has $n = k^4 + t$ with $3t \leq 4k$, and then $\lfloor n^{3/2} \rfloor = k^6 + \frac{3}{2}k^2 t$ exactly. Consequently the fiber of an even cube is full, $G_m = H_m$, and along the fiber of an odd cube the images alternate in parity, so $|2G_m - H_m| \leq 1$.

Proof. If $3t \geq 4k+1$ then $(3k^4+4k+1)^3 \leq (3k^4+3t)^3 = 27(k^4+t)^3$, and $(3k^4+4k+1)^3 \geq 27(k^3+1)^4$, so $n^3 \geq (k^3+1)^4$ and $n \notin \Phi(m)$. Given $3t \leq 4k$, the claimed value $q = k^6 + \frac{3}{2}k^2 t$ satisfies $n^3 - q^2 = \frac{3}{4}k^4t^2 + t^3 \geq 0$ and $(q+1)^2 - n^3 = 2k^6 + 3k^2t + 1 - \frac{3}{4}k^4t^2 - t^3 > 0$, so $q = \lfloor n^{3/2} \rfloor$. For even k the value $k^6 + \frac{3}{2}k^2t$ is even for every t; for odd k the fiber's t are even and the value has the parity of $1 + t/2$. Lean: `cube_fiber_range`, `cube_fiber_sqrt_even`, `cube_fiber_sqrt_odd`, `even_cube_fiber_full`, `odd_cube_fiber_alternating` in `formal/Problems/Juggler/CubeFiber.lean`. ☐

Remark 4.8 (what the window is, arithmetically). $\alpha_m = 0$ exactly means $\frac{3}{2}m^{2/3} \in \mathbb{Z}$, that is $27m^2 = 8k^3$, whose integer solutions are $m = (2e)^3$ and $k = 6e^2$. So the centre of the extreme window is the even cubes of Lemma 4.7, and the window itself is the near-solution set of that equation; the crossings are spaced $\asymp m^{1/3}$ apart, giving $\sim \frac{3}{2}M^{2/3}$ of them below M. Two consequences worth recording. The set is asymmetric about 0, which the symmetric goodness condition of Lemma 4.2 cannot express: that condition discards $\|\alpha_m\| < 22m^{-1/3}$, of width $44m^{-1/3}$, against the $\frac{3}{2}m^{-1/3}$ the extremes actually occupy. Nothing here depends on the difference, since Lemma 4.3 already makes the discarded mass $o(1)$; the point is that the goodness threshold is a convenience, not a boundary of the phenomenon. And the two sides of the window are not symmetric in strength either: full fibers contain the infinite family of Lemma 4.7, proved, whereas an empty fiber requires a near-miss together with a phase condition, and whether there are infinitely many is open.

Remark 4.9 (averaging the rest term). Item 3 of Section 5.1 has coefficient 2/9; a weighted mean even-image share of $1/2 + o(1)$ on that source set would give 1/3. Such a weighted mean is a sufficient additional hypothesis. Conditional equidistribution of the landing phases could help establish it if the approximation errors and the weights were also controlled, but equidistribution is not necessary for one mean identity. Neither Corollary 4.6(1), which averages an independent model phase, nor a pointwise bound on each fiber establishes the required average on an arbitrary backward-closed set. The goodness condition already removes its exceptional fibers at total logarithmic cost $o(1)$.

5. The recursion and the contagion theorem

5.1 Three sources of members of A in $(\sqrt{x}, x]$

Let $x \geq 2$ and $t = \log x$. Three families of members of $A \cap (\sqrt{x}, x]$ are pairwise disjoint.

1. **E-images.** For $m \in A \cap (x^{1/4}, \sqrt{x} - 1]$, $E(m) \subseteq (\sqrt{x}, x]$ (as $m^2 > \sqrt{x}$ and $(m+1)^2 \leq x$). Distinct m give disjoint blocks. By Lemma 3.1 the log-mass is at least $\sum_m \frac{1}{m}(1 - \frac{2}{m}) \geq (1 - 2x^{-1/4})\, \ell_A(x^{1/4}, \sqrt{x} - 1]$, and $\ell_A(x^{1/4}, \sqrt{x} - 1] \geq g_A(t/2) - 2x^{-1/2}$ (at most one integer lies in $(\sqrt{x} - 1, \sqrt{x}]$).

2. **OE-images of E-blocks.** For $m' \in A \cap (x^{3/16}, x^{3/8} - 1]$, $U(m') \subseteq A \cap (\sqrt{x}, x]$ (Lemma 3.2 applied to the even $m \in E(m') \subseteq A$; $I(m') \subseteq (\sqrt{x}, x]$ as $m'^{8/3} > \sqrt{x}$ and $(m' + 1)^{8/3} \leq x$). Distinct m' give disjoint $I(m')$. Each $n \in U(m')$ has $1/n > (m' + 1)^{-8/3}$, so by Proposition 4.4 the log-mass of $U(m')$ is at least $\frac{1}{3} m'^{5/3}(1 - \varepsilon_B(m'))(m' + 1)^{-8/3} \geq \frac{1}{3m'}\big(1 - \varepsilon_B(m') - \frac{8}{3m'}\big)$, and summing, $\geq \frac{1}{3}(1 - \varepsilon'(x))\big(g_A(3t/8) - 2x^{-3/8}\big)$ with $\varepsilon_B(m') = O(m'^{-4/9} \log(m' + 1)) \geq 0$ chosen to bound the error in Proposition 4.4 and $\varepsilon'(x) = \sup_{m' > x^{3/16}}(\varepsilon_B(m') + \frac{8}{3m'}) \to 0$.

3. **OE-images of the rest.** Let $A^E_x = \bigcup_{m'' \in A} E(m'') \cap (x^{3/8}, x^{3/4}]$ be the E-produced part of $A \cap (x^{3/8}, x^{3/4}]$ and A^{rest}_x its complement in $A \cap (x^{3/8}, x^{3/4}]$. The blocks meeting $(x^{3/8}, x^{3/4}]$ come from $m'' \in [x^{3/16} - 1, x^{3/8}]$, so $\ell(A^E_x) \leq \sum_{m''} \frac{m''+1}{m''^2} \leq (1 + 2x^{-3/16})\big(g_A(3t/8) + 2x^{-3/16}\big)$ and $\ell(A^{\mathrm{rest}}_x) \geq g_A(3t/4) - (1 + 2x^{-3/16})\big(g_A(3t/8) + 2x^{-3/16}\big)$. For $m \in A^{\mathrm{rest}}_x$ with $m \leq x^{3/4} - 1$ (this drops at most one element, of log-mass $\leq 2x^{-3/4}$), $\Phi(m) \subseteq (\sqrt{x}, x]$; the fibers are disjoint from each other and from those of item 2 (different m), and for good $m \geq 10^6$ Lemmas 4.2 and 3.2 give log-mass at least $\big(\frac{1}{3} H_m - 2\big)(m+1)^{-4/3} \geq \frac{2}{9m}\big(1 - O(m^{-1/3})\big)$. Bad $m > x^{3/8}$ carry log-mass at most $306x^{-1/8}$ (Lemma 4.3). Hence item 3 contributes at least $\frac{2}{9}(1 - O(x^{-1/8}))\big[\ell(A^{\mathrm{rest}}_x) - 306x^{-1/8} - 2x^{-3/4}\big]_+$.

The images in item 1 are even, those in items 2–3 odd, so the three families are disjoint, and all lie in $(\sqrt{x}, x]$.

5.2 The functional inequalities

Adding items 1 and 2:

$$g_A(t) \;\geq\; (1 - \varepsilon_1(t))\, g_A(t/2) + \tfrac{1}{3}(1 - \varepsilon_2(t))\, g_A(3t/8) - \varepsilon_3(t), \tag{5.1}$$

with $\varepsilon_1 = 2e^{-t/4}$, $\varepsilon_2 = \varepsilon'(e^t)$, $\varepsilon_3 = 2e^{-t/2} + e^{-3t/8}$, all tending to 0. Adding item 3 as well, and noting that the coefficient of the *actual* value $g_A(3t/8)$ is $\frac{1}{3}(1 - \varepsilon_2) - \frac{2}{9}(1 + 2e^{-3t/16}) \to \frac{1}{9} > 0$:

$$g_A(t) \;\geq\; (1 - \varepsilon_1)\, g_A(t/2) + \big(\tfrac{1}{9} - \varepsilon_4\big) g_A(3t/8) + \big(\tfrac{2}{9} - \varepsilon_5\big) g_A(3t/4) - \varepsilon_6(t), \tag{5.2}$$

with $\varepsilon_4, \varepsilon_5, \varepsilon_6 \to 0$ (for t so large that $x^{3/8} \geq 10^6$; the bracket $[\cdot]_+$ may be dropped because the right side of (5.2) is a lower bound for it). Every coefficient of a g_A-value on the right is nonnegative for large t, so lower bounds for g_A at $t/2$, $3t/8$, $3t/4$ transfer.

5.3 The recursion lemma

The passage from a functional inequality with vanishing errors to a power lower bound is a standalone statement.

Lemma 5.1 (recursion lemma). Let $e_1, \dots, e_r \in (0,1)$, $c_1, \dots, c_r \geq 0$, and $\lambda \in (0,1)$ with

$$\zeta := \sum_{i=1}^{r} c_i e_i^{\lambda} - 1 > 0.$$

Put $e_{\min} = \min_i e_i$, $e_{\max} = \max_i e_i$. Let $g : (0,\infty) \to [0,\infty)$ satisfy, for all $t \geq t_1$,

$$g(t) \;\geq\; \sum_{i=1}^{r} \big(c_i - \eta_i(t)\big)\, g(e_i t) - \eta_0(t),$$

where $0 \leq \eta_i(t) \leq c_i$ for $i \geq 1$, $\eta_0(t) \geq 0$, and $\sum_{i \geq 1} \eta_i(t) e_i^{\lambda} \leq \zeta/3$ and $\eta_0(t) \leq \frac{2\zeta}{3} c_0$ for all $t \geq t_1$, for some $c_0 > 0$ with $g(t) \geq c_0$ on $[e_{\min} t_1,\, t_1]$. Then

$$g(t) \;\geq\; K\, t^{\lambda} \quad \text{for all } t \geq e_{\min} t_1, \qquad K = c_0 t_1^{-\lambda}.$$

Proof. On $[e_{\min} t_1, t_1]$, $K t^{\lambda} \leq K t_1^{\lambda} = c_0 \leq g(t)$. Suppose $g(t) \geq K t^{\lambda}$ holds on $[e_{\min} t_1, T]$ for some $T \geq t_1$, and let $t \in (T, T/e_{\max}]$. Then $e_i t \leq e_{\max} t \leq T$ and $e_i t \geq e_{\min} t > e_{\min} t_1$, so each $e_i t$ lies in the interval where the bound is known, and $t \geq t_1$, so

$$\begin{aligned} g(t) &\geq \sum_i (c_i - \eta_i(t)) K (e_i t)^{\lambda} - \eta_0(t) \\ &= K t^{\lambda} \left[\sum_i c_i e_i^{\lambda} - \sum_i \eta_i(t) e_i^{\lambda} \right] - \eta_0(t) \\ &\geq K t^{\lambda} \left(1 + \frac{2\zeta}{3} \right) - \frac{2\zeta}{3} c_0 \geq K t^{\lambda}. \end{aligned}$$

using $c_0 = K t_1^{\lambda} \leq K t^{\lambda}$. Hence the bound holds on $[e_{\min} t_1, T/e_{\max}]$; induction on N covers $[e_{\min} t_1, t_1 e_{\max}^{-N}]$ for every N, i.e. all $t \geq e_{\min} t_1$. $\square$

Lean: `recursion_lemma` in `formal/Problems/Juggler/FateRecursion.lean`, with $e_{\min} \leq e_i \leq e_{\max}$ as bounds rather than extrema and $\lambda > 0$ in place of $\lambda \in (0,1)$; the hypotheses $g \geq 0$ and $\lambda < 1$ are not used.

5.4 The seed

Lemma 5.2 (seed). Every nonempty backward-closed A contains an integer $m \geq 3$, and for any such m,

$$g_A(t) \;\geq\; c_A := \Big(1 - \frac{2}{m^4}\Big)\Big(\frac{0.375}{m+1} - \frac{1}{(m+1)^2 - 1}\Big) > 0 \qquad \text{for all } t \geq 4\log(m+1).$$

Proof. If $A \ni a \leq 2$ then $E(a) \subseteq A$ contains 2 or 4, and $E(2) \ni 4$, so $A \ni 4$. Fix $m \geq 3$ in A and let $S_1 = E(m)$, $S_{k+1} = \bigcup_{m'' \in S_k} E(m'')$. Then $S_k \subseteq A \cap [m^{2^k}, (m+1)^{2^k})$ and, by Lemma 3.1, $\ell(S_{k+1}) \geq (1 - 2m^{-2^k}) \ell(S_k)$, so $\ell(S_k) \geq \ell(S_1) \prod_{j \geq 1} (1 - 2m^{-2^j}) \geq 0.75\, \ell(S_1) \geq \frac{0.375}{m+1}$ for $m \geq 3$. Let $y \geq (m+1)^4$ and choose $k \geq 2$ with $(m+1)^{2^k} \leq y < (m+1)^{2^{k+1}}$. Then $S_k \subseteq (y^{1/4}, y]$ (as $y < m^{2^{k+2}}$). The members of S_k above $\sqrt{y}$ lie in $(\sqrt{y}, y]$; the members in $(y^{1/4}, \sqrt{y} - 1]$ have their E-images in $(\sqrt{y}, y]$, with log-mass at least $(1 - 2m^{-2^k})$ times their own; at most one member lies in $(\sqrt{y} - 1, \sqrt{y}]$, with $1/n \leq 1/((m+1)^2 - 1)$. Hence $g_A(\log y) \geq (1 - 2m^{-4})\big(\ell(S_k) - 1/((m+1)^2 - 1)\big) \geq c_A$, and $c_A > 0$ because $0.375(m+1) \geq 1 + 0.375/(m+1)$ for $m \geq 3$. $\square$

5.5 The theorem

Let λ^* be the root in $(0,1)$ of $2^{-\lambda} + \frac{1}{3}(\frac{3}{8})^\lambda = 1$, let λ_{pair} be the pairing-only root of $2^{-\lambda} + \frac{1}{9}(\frac{3}{8})^\lambda + \frac{2}{9}(\frac{3}{4})^\lambda = 1$, and let λ^{**} be the root of (1.1). All three defining sums are continuous and strictly decreasing on $[0,1]$, with value greater than 1 at 0 and less than 1 at 1; each therefore has a unique root there. Numerically $\lambda^* \approx 0.3774$, $\lambda_{\text{pair}} \approx 0.4480$, the *OEOEE* truncation is 0.4801, the V_3 truncation is 0.4891, the V_4 truncation is 0.4916, the V_5 truncation is 0.4924, and $\lambda^{**} \approx 0.4926$. For comparison: the elementary sweep alone, without Proposition 4.4, gives $2^{-\lambda} + \frac{2}{21}(\frac{3}{4})^\lambda = 1$, root 0.138; perfect fiber equidistribution at this depth would give $2^{-\lambda} + \frac{1}{3}(\frac{3}{4})^\lambda = 1$, root $\lambda_{\text{ideal}} \approx 0.4927$, the ceiling of the depth-two method.

Theorem 5.3 (logarithmic density of a backward-closed set). Let $A \subseteq \mathbb{N}$ be nonempty and backward-closed, and let $0 < \lambda < \lambda^{**}$. There are $c > 0$ and x_0, depending on A and λ, such that

$$\sum_{\substack{n\in A\\ n\le x}} \frac{1}{n} \ \ge\ c\,(\log x)^\lambda \qquad \text{for all } x \ge x_0.$$

The same conclusion for $\lambda < \lambda^*$ uses only Proposition 4.4 (no sweep lemma).

Proof. Apply Lemma 5.1 to $g = g_A$, using the three pairs in (5.2) and the five pairs $(\rho_k, 3^{-(k+1)})$, $2 \le k \le 6$, of (5.10). For $0 < \lambda < \lambda^{**}$, the sum of their $c_i e_i^\lambda$ is greater than 1. Thus $\zeta = \sum_i c_i e_i^\lambda - 1 > 0$. Equations (5.2) and (5.10) have the required form, with the Appendix D errors, all tending to 0. Let $m \ge 3$ be a member of A and $c_0 = c_A$ from Lemma 5.2. Choose t_1 so large that $\frac{729}{8192}t_1 \ge 4\log(m+1)$, that $e^{729t_1/8192} \ge 10^6$, and that for all $t \ge t_1$ the errors satisfy $\sum_i \eta_i(t)e_i^\lambda \le \zeta/3$ and $\eta_0(t) \le \frac{2\zeta}{3}c_A$. Lemma 5.2 gives $g_A \ge c_A$ on $[\frac{729}{8192}t_1, t_1]$. The first induction step of Lemma 5.1 covers up to $t_1/(\frac{3}{4}) = \frac{4}{3}t_1$, at which $\frac{729}{8192}t = \frac{243}{2048}t_1$. Lemma 5.1 then gives $g_A(t) \ge Kt^\lambda$ for all $t \ge \frac{729}{8192}t_1$, and $\sum_{n\in A,\, n\le x} 1/n \ge g_A(\log x)$. For $\lambda < \lambda^*$ use (5.1) with $(e_i, c_i) = (\frac{1}{2}, 1), (\frac{3}{8}, \frac{1}{3})$. $\square$

Corollary 5.4 (natural density, infinitely often). For every $\lambda < \lambda^{**}$ there is $c > 0$ such that for every $X \ge x_0$ some $y \in (\sqrt{X}, X]$ satisfies $\#\big(A \cap (y/2, y]\big) \ge c\,y\,(\log y)^{\lambda-1}$.

Proof. $\ell_A(\sqrt{X}, X] = g_A(\log X) \ge K(\log X)^\lambda$ is spread over at most $\frac{1}{2}\log_2 X + 2$ dyadic blocks $(y/2, y]$ with $\sqrt{X} < y \le X$; one of them has $\ell_A(y/2, y] \ge K'(\log X)^{\lambda-1}$, hence $\#(A \cap (y/2, y]) \ge \frac{y}{2}\,\ell_A(y/2, y] \ge \frac{K'}{2}\,y\,(2\log y)^{\lambda-1}$, because $\log X \le 2\log y$ and $\lambda - 1 < 0$. $\square$

Corollary 5.5 (fate contagion). Let φ be a fate realized by at least one start: the trivial cycle, a particular nontrivial cycle C, or divergence. Then $\{n : \text{fate}(n) = \varphi\}$ satisfies Theorem 5.3 and Corollary 5.4. In particular:

1. $\sum_{n\in R,\ n\le x} 1/n \gg (\log x)^\lambda$ for every $\lambda < \lambda^{**}$.
2. If some start does not reach 1, then $\sum_{n\notin R,\ n\le x} 1/n \gg (\log x)^\lambda$, and on infinitely many dyadic blocks the failures have natural density $\gg (\log y)^{\lambda-1}$.
3. If a nontrivial cycle exists, the set of starts that enter it has the same lower bounds; if a divergent orbit exists, so does the set of divergent starts.

Proof. Lemma 2.1 and Theorem 5.3. $\square$

5.6 Remarks

The status of stronger density bounds. The theorem gives a logarithmic counting lower bound. It does not imply positive natural or logarithmic density. The ideal model of Section 5.7 has critical exponent 1, but the corresponding arithmetic hypotheses are not proved. Corollary 5.4 gives a natural-density lower bound on infinitely many blocks. No pointwise lower bound on every dyadic block is proved here. The even-only tree of a single seed is lacunary, but it need not be backward-closed under the full Juggler map and therefore is not a counterexample for that larger class.

Sharpness of the constants. The monotone pairing gives $\frac{1}{3}H_m - 2$; the observed floor on good fibers is 0.328 at $\alpha_m \approx \frac{1}{3}$ (Section 11). The remaining depth-two gap $0.448 \to 0.4927$ is a dynamical averaging problem for the low-even set $P = \{m : G_m/H_m \le 0.40\}$. The nominal exponent ρ_w gives a useful scale diagnostic for candidate productions, but Model calculation 5.13 shows that it does not collapse the actual nested fiber to one monomial interval. The finite V_k constructions below use their own exact nested fibers; their limiting transfer-matrix value 0.4927 is a model ceiling, not a proved infinite-family exponent.

What is excluded. Nothing. Corollary 5.5 does not say that a cycle or a divergent orbit is impossible; it says that either would be common.

5.7 Finite productions and comparison models

Every production used above is a parity word w applied backwards. Write $a = \#\{E \text{ in } w\}$, $b = \#\{O \text{ in } w\}$. The source of a w-production sits at log-scale $\rho_w t$ with

$$\rho_w = 2^{-a}\left(\tfrac{3}{2}\right)^b, \qquad c_w^{\text{ideal}} = \frac{2^{-|w|}}{\rho_w}, \tag{5.7}$$

the second being the log-mass coefficient when the fiber realizes the fair share $2^{-|w|}$. Formula (5.7) returns every coefficient in this section and in Appendix C: $c_E = 1$, $c_{OE} = c_{OEE} = \frac{1}{3}$, $c_{OOEEE} = \frac{1}{9}$.

Proposition 5.12 (an ideal-model critical exponent). For $0 < \eta \le 1$, define $\lambda(\eta)$ as the root in $(0, 1]$ of the model characteristic equation

$$2^{-\lambda} + \tfrac{\eta}{3}\left(\tfrac{3}{2}\right)^{\lambda} = 1. \tag{5.8}$$

Then $\lambda(1) = 1$ and $\lambda'(1) = 1/\ln(4/3) = 3.4760\ldots$. This is an algebraic model, not a Juggler counting theorem.

Proof. On $[0, 1]$, the left side is strictly decreasing: its derivative is at most $-\frac{1}{2}\ln 2 + \frac{1}{2}\ln(3/2) < 0$. At $\lambda = 0$ it is $1 + \eta/3 > 1$, and at $\lambda = 1$ it is $(1 + \eta)/2 \le 1$. This gives a unique root in that interval, equal to 1 exactly for $\eta = 1$. Implicit differentiation at $(1, 1)$ gives the derivative. At $\eta = 1$ the two tilted weights are both 1/2, the fair-coin path weights. They are not both 1/2 for general η. □

The root at 1 describes what the ideal characteristic equation permits. Exact nested fibers and their shares must still be proved before a proposed production list yields an arithmetic exponent.

Model calculation 5.13 (collapsed-power fiber; not an identity for the Juggler map). If one replaces the nested branch belonging to a production word w by the single model map $\widetilde{J}_w(n) = \lfloor n^{\rho_w} \rfloor$, then its fiber over a source m' is exactly $I_{\text{mod}}(m') = [m'^{1/\rho_w}, (m'+1)^{1/\rho_w})$, and at scale $P \asymp m'^{1/\rho_w}$

$$|I_{\text{mod}}(m')| = \tfrac{1}{\rho_w}\, m'^{1/\rho_w - 1}(1 + O(1/m')) \asymp P^{1-\rho_w}.$$

The corresponding assertion for the exact map is false. For example, 1015 follows

$$1015 \xrightarrow{O} 32336 \xrightarrow{E} 179 \xrightarrow{O} 2394 \xrightarrow{E} 48 \xrightarrow{E} 6,$$

so its word is $OEOEE$ and $J^5(1015) = 6$. But $7^{32} \le 1015^9 < 8^{32}$, hence $\lfloor 1015^{9/32} \rfloor = 7$, not 6. Nested floors cannot in general be replaced by one final floor.

Appendix C's localized parity estimate requires an *actual proved* parameter interval of length at least $P^{1/2}$. Thus $\rho_w \le \frac{1}{2}$ is only the collapsed model's candidate threshold; it neither proves the needed interval nor the ideal share. The words E, OEE, and conditionally $OOEEE$ have separate fiber arguments in their cited results. The nominal lengths $P^{1/2}, P^{5/8}, P^{23/32}$ remain useful bookkeeping, while the nominal OE and $OOEE$ lengths $P^{1/4}$ and $P^{7/16}$ only screen where that particular localization estimate could apply.

The conditional run model and the K_3 interface. Production words must be prefix-free. First passage of the nominal ρ_w below $\frac{1}{2}$ gives a candidate list, not a theorem that its fibers have the model length or ideal share. A run of r consecutive O's forces r nested 3/2-powers plus the closing square root — Paper B at depth $r+1$. Paper B is complete to depth 4, and to depth 5 except $OOOO*$, which is the parked kernel K_3. Under the additional hypothesis that every required candidate production through run length r is established at the stated share, the transfer matrix gives:

r	Paper B depth	conditional $\lambda(r)$, pairing share	conditional $\lambda(r)$, ideal-share model
1	2	0.4480	0.4927
2	3	0.6247	0.7180
3	4	0.7095	0.8414
4	5	0.7516	0.9121

The table is a price list of hypothetical inputs, not an attained ladder. Appendix C establishes one $r = 2$ word conditionally and prints 0.5392; nothing here supplies the other words needed for 0.6247 or 0.7180. Row $r = 4$ would also meet the parked K_3 kernel, but the model does not prove that this is the only obstruction.

The finite productions used in Theorem 1. Put $V_k = (OE)^{k-1}OEE$, for $1 \le k \le 6$, and $\rho_k = \frac{1}{2}(3/4)^k$. These six words are prefix-free. The exact landing windows have iterated ceiling endpoints; they are not the single-power windows of Model calculation 5.13. Appendix D proves, for each of these fixed words, the needed two-sided asymptotic and the resulting logarithmic coefficient $c_k = 3^{-k}$.

The V_k-starts for $k \ge 2$ are contained in family 3 of Section 5.1: their two-step images are odd V_{k-1}-produced members of A. Before counting them at their improved share, remove those source sets from family 3. The old contribution is $(2/9)c_{k-1}$; the new one is c_k. Their difference is

$$c_k - \tfrac{2}{9}c_{k-1} = 3^{-(k+1)}, \qquad 2 \le k \le 6. \tag{5.9}$$

Disjointness and the two-sided estimates justify making all five replacements simultaneously. Thus (5.2) gains

$$\sum_{k=2}^{6}\bigl(3^{-(k+1)} - o(1)\bigr)g_A(\rho_k t) - o(1). \tag{5.10}$$

The corresponding roots are 0.4801, 0.4891, 0.4916, 0.4924, and 0.4926. This is the finite input used in Theorem 5.3.

For comparison only, summing the formal infinite series in (5.9) gives the ideal depth-two equation $2^{-\lambda} + \frac{1}{3}(3/4)^{\lambda} = 1$, with root 0.4927. Neither a uniform estimate in k nor attainment of that endpoint is asserted. Adding just V_2 to the conditional construction in Appendix C would give root 0.5665; the analogous infinite model gives 0.5769. These comparisons are not the exponents claimed in Theorems 1 or C.5.

6. Odd generation and the exact first-letter decomposition

Write $S = \{\lfloor m^{3/2}\rfloor : \ m \text{ odd}\}$ for the image of the odd integers, and S_{odd} for its odd members. The set S is sparse: it has $\asymp z^{2/3}$ members below z.

6.1 Odd generation

Theorem 6.1 (odd generation; Lean). Let A be forward- and backward-closed with $1 \notin A$. Then:

1. every $n \in A$ descends by even steps only to an odd member $m \geq 3$ of A (`exists_odd_ancestor_ge_three`);
2. for odd n, $n \in A \iff \lfloor n^{3/2} \rfloor \in A$ (`odd_mem_iff`), so the odd members of A are exactly the odd preimages of $A \cap S$;
3. $A \neq \emptyset \iff A \cap S \neq \emptyset$, i.e. A contains the image of some odd $m \geq 3$ (`nonempty_iff_odd_image_mem`).

Proof. (1) If $n \in A$ is even, $\lfloor \sqrt{n} \rfloor \in A$ by forward closure and is smaller; the chain stops at an odd integer, which is ≥ 3 since $1 \notin A$ and $2 \notin A$ ($J(2) = 1$). (2) Forward closure gives one direction, backward closure the other. (3) follows from (1) and (2). $\square$

Applied to the failure set F: every positive integer reaches 1 iff no image $\lfloor m^{3/2} \rfloor$ of an odd m fails to reach 1, and F is the E-forest over the odd preimages of $F \cap S$. The theorem changes the geometry of the problem. An arbitrary failure set is replaced by a set of *seeds* — odd images, a set of density $\asymp z^{-1/3}$ at scale z — together with the deterministic even ancestry of each odd preimage (its E-forest, whose levels are the intervals of Lemma 3.1) and the unique odd preimage of each seed in S. Two consequences are used below: the log-count of F is controlled by that of its odd members (Theorem 7.2), and the first-letter decomposition of F has exactly one term that the productions do not control (Section 6.2).

6.2 The exact decomposition and the free term

Let A be forward- and backward-closed. On $(\sqrt{x}, x]$ its members split by first letters into three exact pieces (Figure 3):

- the **even** members, $= \bigsqcup_{m \in A} E(m) \cap (\sqrt{x}, x]$, of log-mass $\ell_A(x^{1/4}, \sqrt{x}] + O(x^{-1/4})$ (Lemma 3.1);
- the *OE*-**type** odd members ($\lfloor n^{3/2} \rfloor$ even), $= \bigsqcup_{m \in A} \{n \in \Phi(m) \cap (\sqrt{x}, x] : \lfloor n^{3/2} \rfloor \text{ even}\}$ (Lemma 3.2), of log-mass between $\frac{2}{9}$ and $\frac{4}{9}$ times $\ell_A(x^{3/8}, x^{3/4}]$ on good fibers (Lemma 4.2, two-sided pairing) and $\frac{1}{3}(1 + o(1))$ times it on E-blocks (Proposition 4.4);
- the *OO*-**type** odd members ($\lfloor n^{3/2} \rfloor$ odd), $= \{n(m) : m \in A \cap S_{\text{odd}},\ \sqrt{x} < n(m) \leq x\}$, of log-mass $\sum_{m \in A \cap S_{\text{odd}},\ \sqrt{x} < n(m) \leq x} 1/n(m)$, where $n(m)$ is the unique odd preimage. Writing the cutoff on $n(m)$ keeps the floor endpoints exact.

Let $\varphi_A(t) = \ell_A(e^{t/2}, e^t]/(t/2)$ be the logarithmic density of A on $(\sqrt{x}, x]$, $t = \log x$. Define $\varphi_A^{\text{fib}}(3t/4)$ by the normalization $\frac{1}{4} \cdot \frac{t}{2} \cdot \varphi_A^{\text{fib}}(3t/4) =$ log-mass of the *OE*-type members, and $\psi_A(t)$ by $\frac{1}{4} \cdot \frac{t}{2} \cdot \psi_A(t) =$ log-mass of the *OO*-type members. Then $\varphi_A^{\text{fib}}(s)$ is the density of A on $(e^{s/2}, e^s]$ weighted, on each fiber, by twice the even-image fraction $p_m = G_m/H_m$ of that fiber — so the weighted and unweighted masses agree to vanishing relative error on complete E-block unions, with an additional exponentially small boundary error (Proposition 4.4), and $\varphi_A^{\text{fib}}(s) \leq \frac{4}{3}\varphi_A(s) + O(e^{-s/6})$ always (Lemmas 4.2, 4.3) — and $\psi_A(t)$ is the log-weighted share of A among the *OO*-type odd integers of $(\sqrt{x}, x]$, up to a factor $1 + O(e^{-\delta t})$ from the depth-two equidistribution of the parity of $\lfloor n^{3/2} \rfloor$. With these definitions the three pieces give the identity with boundary error

$$\varphi_A(t) = \tfrac{1}{2}\varphi_A(t/2) + \tfrac{1}{4}\,\varphi_A^{\text{fib}}(3t/4) + \tfrac{1}{4}\,\psi_A(t) + O(e^{-t/4}/t), \tag{6.1}$$

whose only error is the boundary term of Lemma 3.1. For $A = \mathbb{N}$ every term is $1 + o(1)$. For $A = F$ the boundary condition is $\varphi_F(t) = 0$ for $t \leq \log N_0$.

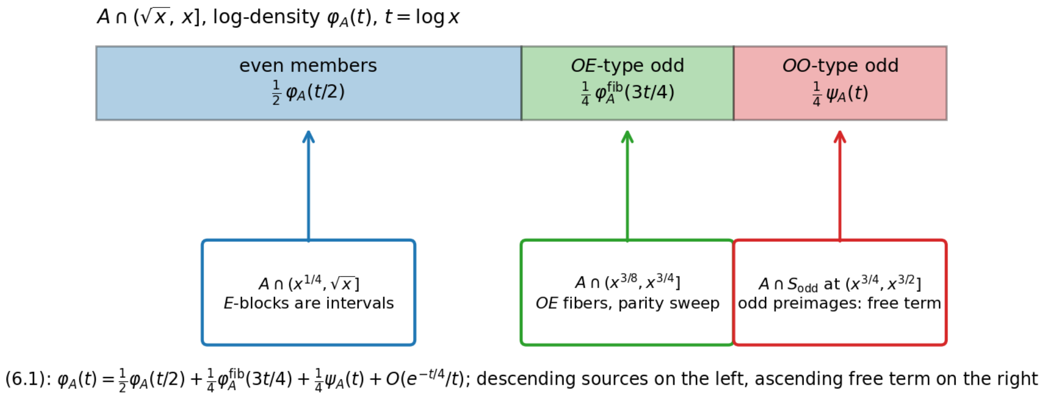

Figure 3: The first-letter decomposition of a two-way closed set on $(\sqrt{x}, x]$: the even members come from scale $x^{1/2}$ through E-blocks, the OE-type odd members from scale $x^{3/4}$ through fibers, and the OO-type odd members are the odd preimages of the odd images of A at scale $x^{3/2}$ — the free term.

Definition 6.2 (S-fairness). A two-way closed set A is *S-fair at scale t* if $\psi_A(t) \leq \varphi_A(3t/2)$, i.e. if A is not over-represented among the odd images of odd integers at scale $e^{3t/2}$ relative to its density among all integers there; it is *S-fair* if this holds for all $t \geq t_0$.

Proposition 6.3 (exact statements). (i) $\psi_F \equiv 0$ iff $F = \emptyset$: if $F \neq \emptyset$, its minimum is odd with odd image, so $\psi_F(t) > 0$ on a shell containing $\min F$. (ii) The identity (6.1) holds for F with the stated error, and its right side contains exactly one term, $\frac{1}{4}\psi_F(t)$, that is not determined by F at smaller scales through the two productions.

Proof. (i) If $F = \emptyset$ then $\psi_F \equiv 0$. If $F \neq \emptyset$, let $n_F = \min F$. It is odd (an even failure has the smaller failure $\lfloor\sqrt{n_F}\rfloor$), and its image is odd (if $J(n_F)$ were even, $J^2(n_F) = \lfloor n_F^{3/4} \rfloor < n_F$ would be a smaller failure). So n_F is an OO-type failure and $\psi_F(t) > 0$ for t with $e^{t/2} < n_F \leq e^t$. (ii) is the decomposition. ☐

Lean: `minimal_failure_odd_odd` and `exists_minimal_failure` (the least failure is odd with odd image), and the three pieces as `first_letter_trichotomy` with `first_letter_pieces_disjoint`, in `formal/Problems/Juggler/FateFirstLetter.lean`; the log-mass bookkeeping of (6.1) is not formalized.

Remark 6.4 (the walk heuristic; not a theorem). If F were S-fair, if the fiber weights were ideal ($\varphi_F^{\mathrm{fib}} = \varphi_F$), and if the error term of (6.1) were zero, then (6.1) would give $\varphi_F(t) \leq \frac{1}{2}\varphi_F(t/2) + \frac{1}{4}\varphi_F(3t/4) + \frac{1}{4}\varphi_F(3t/2)$: the harmonic inequality of a random walk on $\log t$ with steps $\log\frac{1}{2}, \log\frac{3}{4}, \log\frac{3}{2}$ of probabilities $\frac{1}{2}, \frac{1}{4}, \frac{1}{4}$, whose drift -0.317 is negative, so that the boundary condition $\varphi_F = 0$ below $\log N_0$ would force $\varphi_F \equiv 0$. None of the three conditions is available with the constants proved here: the fiber weights are only known to lie in $[\frac{2}{3}, \frac{4}{3}]$, so the coefficients of the inequality can sum to $\frac{1}{2} + \frac{1}{3} + \frac{1}{4} = \frac{13}{12} > 1$, and the boundary error $O(e^{-t/4}/t)$ is not zero at the scales just above the floor where the walk exits. We therefore do not claim that S-fairness implies the conjecture. What is exact is Proposition 6.3; what is proved about the free term is Proposition 10.1, which identifies $\psi_F(t)$ with a quantity bounded by every hypothesis of Sections 8–9.

6.3 Where Papers A and B sit

Paper A [11] constrains the fate Lachesis from the inside: finance and the walk charge bound the *states* of a hypothetical cycle (minimum $> 3.5 \cdot 10^8$, period ≥ 780239). Theorem 1 constrains the basin: if the cycle exists, its basin is a two-way closed class with the cycle's states as seeds and log-count $\gg (\log x)^{\lambda}$ for every $0 < \lambda < \lambda^{**}$. In the present argument the two estimates do not meet: finance bounds the seed, contagion the growth from the seed, and no inequality bounds a basin from above. Paper B [12] controls the descending branches of (6.1) — the fairness of the landing distributions of E, OE and, with its depth-4 and depth-5 theorems, of deeper contracting words — on dyadic blocks; Appendix C uses one statement of that type on sub-dyadic intervals as an explicit hypothesis. Neither touches ψ_F: the ascending branch OO sends mass to $x^{3/2}$, and its return is the nested-floor parity at all depths.

7. The almost-all equivalence

Corollary 7.1 (logarithmic form). The following are equivalent: (1) every $n \ge 1$ reaches 1; (2) for some $\lambda < \lambda^{**}$, $\sum_{n\le x,\ n\notin R} 1/n = o((\log x)^\lambda)$; (3) for some $0 < \lambda < \lambda^{**}$, the starts $n \le x$ whose orbit never enters $[1, N_0]$ have logarithmic count $o((\log x)^\lambda)$.

Proof. (1)⇒(3): the set is empty. (3)⇒(2): an orbit that enters $[1, N_0]$ reaches 1, so F is contained in the set of (3). (2)⇒(1): F is backward-closed; if it were nonempty, Theorem 5.3 would contradict (2). □

Theorem 7.2 (a Tao-type bound with rate implies the conjecture). Suppose that for some $e > 1 - \lambda^{**} \approx 0.5074$ and all sufficiently large y,

$$\#\{n \text{ odd},\ y < n \le 2y :\ n \notin R\} \ \le\ \frac{y}{(\log y)^e}.$$

Then $R = \mathbb{N}$.

Proof. By decreasing e if necessary, assume $1 - \lambda^{**} < e < 1$; the hypothesis is preserved for large y. Suppose $F \ne \emptyset$. By Theorem 6.1 every $n \in F$ lies in the E-tree of an odd member of F. For an odd $n_0 \in F$ the level-j set $S_j(n_0)$ of its E-tree has log-mass at most $\prod_{i<j}(1+n_0^{-2^i})/n_0 \le 2/n_0$ (Lemma 3.1 gives $\sum_{n\in E(m)} 1/n \le (m+1)/m^2$), and only levels with $n_0^{2^j} \le x$, i.e. $j \le \log_2 \log x$, meet $[1, x]$. Hence

$$\begin{aligned}
\sum_{\substack{n\in F\\ n\le x}} \frac{1}{n} &\le 2(1+\log_2\log x) \sum_{\substack{n_0\in F \text{ odd}\\ n_0\le x}} \frac{1}{n_0} \\
&\le 2(1+\log_2\log x)\left(\sum_{k_0\le k\le \lfloor \log_2 x\rfloor} (k\ln 2)^{-e} + O(1)\right) \\
&\ll (\log x)^{1-e}\log\log x.
\end{aligned}$$

By Theorem 5.3 the left side is $\ge K(\log x)^\lambda$ for every $\lambda < \lambda^{**}$ and all large x. Choosing $\lambda \in (1-e, \lambda^{**})$ gives a contradiction. □

Theorem 7.3 (equivalence). Every positive integer reaches 1 if and only if there is $e > 1 - \lambda^{**}$ such that $\#\{n \text{ odd} \in (y, 2y] :\ n \notin R\} \le y(\log y)^{-e}$ for all large y.

Proof. If every integer reaches 1 the left side is 0. Conversely Theorem 7.2. □

The threshold $1-\lambda^{**}$ is the complement of the contagion exponent; with λ^{***} of Appendix C it becomes 0.4608. Any improvement of the contagion exponent lowers the rate required of the almost-all statement, and $\lambda \to 1$ would make any positive rate suffice — but that improvement requires all descent certificates and is circular here (Section 5.6).

7.1 Comparison with the Collatz map: a structural analogy

The comparison with Tao's theorem [6] is a structural analogy, not a transfer of methods, and it concerns two specific features.

Depth. Juggler's multiplicative changes of $\log n$ motivate a walk on $\log\log n$ and the depth $O(\log\log y)$. This is the depth used by the sufficient hypotheses here, not a proved stopping-time bound for every orbit. For the accelerated Collatz map of Terras and Everett, parity words of length k correspond to residue classes modulo 2^k. Their counts in an interval of length y have an additive $O(1)$ boundary error; relative control by that fact alone becomes ineffective once 2^k is comparable to or larger than y. Saying that every depth of order $\log y$ has singleton cylinders would be too strong: the multiplicative constant in the depth matters.

Preimage counts. The result of Krasikov–Lagarias [7] cited in Section 1.3 has a different counting scale from Theorem 1. An $x^{0.84}$ lower bound is compatible with a bounded reciprocal sum, and by itself cannot contradict a logarithmic error bound. This comparison explains why the particular argument of Theorem 7.2 does not transfer. It does not exclude other almost-all-to-all arguments for Collatz.

Tao's theorem [6] has an arbitrarily slowly growing target and logarithmic-density exceptional set. Our eventual-entry equivalence has a fixed verified target and a specified dyadic rate; obtaining that rate remains open. No part of Tao's renewal argument is claimed to have been supplied for Juggler.

8. From parity control to the almost-all bound

8.1 Notation

For a word $w \in \{O, E\}^d$ and $t \leq d$ write $o_t(w)$ for the number of odd letters among the first t and

$$u_t(w) = o_t(w) \log_2 3 - t$$

for the *exponent walk*, so that the ideal image after t letters is $n^{2^{u_t}}$. For $y > N_0$ put

$$L(y) = \log_2 \frac{\log 2y}{\log N_0}, \qquad d(y) = \lceil C\, L(y) \rceil,$$

with a constant $C \geq 5$ fixed below. A word w of length d is *L-bad* if $u_t(w) > -L$ for every $1 \leq t \leq d$ (its walk never descends to $-L$). $\mathrm{word}_d(n)$ is the realized itinerary of the first d steps of n (letter s is the parity of $J^{s-1}(n)$); the *cylinder* $[w]_y$ is the set of odd $n \in (y, 2y]$ with $\mathrm{word}_{|w|}(n) = w$. Odd starts have first letter O, so the fair share of an O-rooted cylinder of depth d relative to the reference mass $N = y/2$. The actual number of odd starts in $(y, 2y]$ is $N+O(1)$; this bounded endpoint error is absorbed in the asymptotic bounds below is $2^{-(d-1)}N$.

8.2 Envelope descent and the rarity of bad words

Lemma 8.1 (envelope descent). Let $n \in (y, 2y]$ and $d \geq 1$. If $\mathrm{word}_d(n)$ is not $L(y)$-bad, then $n \in R$.

Proof. Some $t \leq d$ has $u_t \leq -L(y)$, i.e. $3^{o_t}/2^t \leq \log N_0/\log 2y \leq \log N_0/\log n$, i.e. $n^{3^{o_t}} \leq N_0^{2^t}$ as integers. The power envelope $J^t(n)^{2^t} \leq n^{3^{o_t}}$ (Paper A, Theorem 2.2; `power_bound_word`) gives $J^t(n) \leq N_0$, so $J^t(n) \in R$ and $n \in R$ by backward closure. Lean: `iterate_le_of_envelope`, `mem_of_envelope_floor`, `reachesOne_of_itinerary_envelope`. □

Lemma 8.2 (Chernoff). For $C \geq 5$ put

$$p_C = \frac{1 - 1/C}{\log_2 3}, \qquad e(C) = \frac{C\, D(p_C \,\|\, \frac{1}{2})}{\ln 2}, \qquad D(p\|\tfrac{1}{2}) = p \ln(2p) + (1-p)\ln(2(1-p)).$$

For $L > 0$ and $d \geq CL$, the number of L-bad words of length d is at most $2^d \cdot 2^{-e(C)L}$. Moreover, for every $\varepsilon > 0$ there is L_0 such that for $L \geq L_0$ and $d = \lceil CL \rceil$ the number of L-bad words of length d beginning with O is at most $2^{d-1} \cdot 2^{-(e(C)-\varepsilon)L}$.

Proof. An L-bad word has $u_d > -L$, i.e. $o_d > (d-L)/\log_2 3 \geq p_C d$ (as $L/d \leq 1/C$). For a fair coin, $\Pr[\mathrm{Bin}(d, \frac{1}{2}) \geq pd] \leq e^{-dD(p\|1/2)}$ for $p \geq \frac{1}{2}$, and $D(\cdot\|\frac{1}{2})$ is increasing on $[\frac{1}{2}, 1]$; $p_C \geq \frac{1}{2}$ for $C \geq 5$. Hence the count is at most $2^d e^{-dD(p_C\|1/2)} \leq 2^d e^{-CL\, D(p_C\|1/2)} = 2^d 2^{-e(C)L}$. For the

O-rooted count, the remaining $d-1$ letters contain $o_d - 1 > (d-L)/\log_2 3 - 1 = p'(d-1)$ odd letters with $p' = p_C - O(1/L)$; the same bound with $d-1$ trials gives $2^{d-1}e^{-(d-1)D(p'\|1/2)}$, and $(d-1)D(p'\|1/2) \ge (e(C)-\varepsilon)L\ln 2$ for L large. □

Lean: the first statement, in the exact form $\#\{w\ L\text{-bad}, |w| = d\} \le 2^d\, 2^{-e(C)L}$ for $C \ge 5$, $d \ge CL$, $d \ge 1$, is `LBad_count_le` in `formal/Problems/Juggler/FateChernoff.lean`: the Markov tilt `weight_markov` on the constant weight (`weightGen_one`, $(1+x)^d$), the value $\exp(d\,h(p))$ at $x = p/(1-p)$ (`tilt_value`, `count_oddCount_ge_le_exp`), Gibbs' inequality $D(p\|\frac{1}{2}) \ge 0$ (`klHalf_nonneg`), and $p_C \in [\frac{1}{2}, 1)$ from $\log_2 3 \le 8/5$ (`half_le_pC`, `pC_lt_one`). The O-rooted statement with ε stays a human proof.

Numerically $e(18) \approx 0.480$, $e(19) \approx 0.527$, $e(20) \approx 0.574$, $e(21) = 0.621$, $e(25) = 0.812$, $e(30) = 1.054$; $e(C)$ grows linearly in C with slope $D(1/\log_2 3 \,\|\, \frac{1}{2})/\ln 2 = 0.050$.

8.3 The cylinder hypothesis and the bound

Hypothesis $\mathrm{H}(C, A)$ **(log-log-depth cylinder bound).** For all sufficiently large y, with $d = d(y)$, every word $w \in \{O, E\}^d$ that begins with O and is $L(y)$-bad satisfies

$$\#[w]_y \;\le\; 2^{-(d-1)} \cdot \frac{y}{2} + \frac{y}{(\log y)^A}.$$

Only an upper bound is asked, only at one depth per scale, and only for the $L(y)$-bad words. Since $2^{-(d-1)}y/2 \asymp y(\log y)^{-C}$, $\mathrm{H}(C, A)$ with $A > C$ says that no O-rooted cylinder of depth $d(y)$ whose envelope has not crossed the floor exceeds its fair share by more than a relative $O((\log y)^{C-A})$. The restriction is essential: the older all-word version is false because already absorbed starts can share a long terminating tail and overpopulate a cylinder without contributing any live mass; see the absorbed-cylinder obstruction. This obstruction does not refute $\mathrm{H}(C, A)$ as stated here.

Theorem 8.3 (Tao-type bound from the cylinder hypothesis). Assume $\mathrm{H}(C, A)$ with $C \ge 5$ and $A > C + e(C)$. Then for every $\varepsilon > 0$ and all sufficiently large y,

$$\begin{aligned}\#\{n \text{ odd in } (y, 2y] : n \notin R\} &\le \#\{n \text{ odd in } (y, 2y] : J^t(n) > N_0\ \forall t \le d(y)\}\\ &\le \frac{y}{2}\left(\frac{\log(2y)}{\log N_0}\right)^{-(e(C)-\varepsilon)}.\end{aligned}$$

In words: all but $O(y(\log y)^{-e(C)+\varepsilon})$ odd starts in $(y, 2y]$ enter $[1, N_0]$ within $d(y) = O(\log\log y)$ steps.

Proof. A failure never enters the verified floor, giving the first inequality. A live start has a bad word by Lemma 8.1. Apply Lemma 8.2 with $\varepsilon/2$ in place of ε. Every odd start has an O-rooted word; by Lemma 8.2 the O-rooted bad words number at most $2^{d-1}2^{-(e(C)-\varepsilon/2)L(y)}$, and by $\mathrm{H}(C, A)$ each carries at most $2^{-(d-1)}y/2 + y(\log y)^{-A}$ odd starts. Hence the count is at most

$$\frac{y}{2}\, 2^{-(e(C)-\varepsilon/2)L(y)} + 2^{d-1}\,\frac{y}{(\log y)^A} \;\le\; \frac{y}{2}\Big(\frac{\log 2y}{\log N_0}\Big)^{-(e(C)-\varepsilon/2)} + \Big(\frac{\log 2y}{\log N_0}\Big)^C \frac{y}{(\log y)^A},$$

using $2^{d-1} \le 2^{CL}$. The second term is $O(y(\log y)^{C-A}) = o(y(\log y)^{-e(C)})$ because $A > C + e(C)$. For large y, both terms fit inside the displayed bound with exponent $e(C) - \varepsilon$, using the remaining $\varepsilon/2$ slack. □

Lean: the exact skeleton, in `formal/Problems/Juggler/FateChernoff.lean`. `oddFailures_subset_bad_cylinders` is Lemma 8.1 in covering form: with every start up to N_0 reaching 1, an odd $n \in (y, 2y]$ that does not reach 1 lies in the cylinder of a word whose every prefix fails the integer envelope comparison $N_0^{2^t} < (2y)^{3^{o_t}}$ (`EnvelopeBad`); `LBad_of_envelopeBad` identifies those words as $L(y)$-bad with $L(y) = \log_2(\log 2y/\log N_0)$; `oddFailures_card_le` is the union bound, and `oddFailures_card_le_chernoff` composes it with Lemma 8.2: if every $L(y)$-bad cylinder of depth $d \geq CL(y)$ holds at most M starts, the odd failures in $(y, 2y]$ number at most $2^d\, 2^{-e(C)L(y)} M$. `oddFailures_card_le_explicit` substitutes $d = \lceil CL(y) \rceil$ and the bound of $\mathrm{H}(C, A)$ at the scale y, for O-rooted words only (`cylinder_even_root_empty`): with $\Lambda = \log 2y / \log N_0$, the odd failures in $(y, 2y]$ number at most $y\,\Lambda^{-e(C)} + 2\Lambda^C y(\log y)^{-A}$, at every $y \geq 2$ and without ε. The displayed form follows by absorbing the factor 2 into Λ^{ε} and the second term into the first for $A > C + e(C)$; that absorption is not formalized. Corollary 8.4 uses only the exponent.

Corollary 8.4 (the conjecture from a cylinder bound). If $\mathrm{H}(C, A)$ holds for some $C \geq 19$ and $A > C + e(C)$, then every positive integer reaches 1. Under the conditional exponent λ^{***} of Appendix C the same conclusion holds for $C \geq 18$.

Proof. Theorem 8.3 gives the hypothesis of Theorem 7.2 with $e = e(C) - \varepsilon \geq e(19) - \varepsilon > 1 - \lambda^{**}$; with λ^{***} the threshold is $0.4608 < e(18) \approx 0.480$. The pairing-only intermediate still needed $C \geq 20$ ($e(20) = 0.574 > 0.5520$). □

8.4 Constants

The least integer depth constant in this sufficient criterion is $C = 19$ ($e(19) \approx 0.527 > 1 - \lambda^{**}$). With the certified floor $N_0 = 3.5 \cdot 10^8$, the table below keeps $C = 20$ as a conservative a-fortiori display:

y	$L(y)$	$d(y)$	exact fair-coin bad probability	$(\log y)^{-0.6}$	least depth for rate 0.6
10^{20}	1.25	25	0.065	0.100	19
10^{100}	3.55	72	0.017	0.038	56
10^{1000}	6.87	138	0.0038	0.0096	117

The bad probability is the fair-coin probability that a walk of length $d(y)$ never reaches $-L(y)$, by dynamic programming over the walk (unconditioned first letter; conditioning on $u_1 = \log_2 3 - 1$ roughly doubles the finite-depth values and leaves the exponent unchanged); the last column is the least depth at which it drops below $(\log y)^{-0.6}$ — about $17\,L(y)$. With the Lean floor $N_0 = 260$ the depths grow by about 38 letters. For comparison, Paper B controls depth 4 (all words) and two words of depth 5, with relative error $y^{-1/96}$ on dyadic blocks; the hypothesis asks for the bad cylinders at depth $d(y) \to \infty$ with relative error $o(1)$ — weaker in rate than a power saving, unbounded in depth.

9. Weaker forms of the hypothesis

Theorem 8.3 asks for a fair-share upper bound on every envelope-bad cylinder at one depth. Lower bounds and control of cylinders after their envelope has crossed the floor are not needed. This section gives the sufficient conditions, including a single moment bound, and records what that form does not need.

9.1 One-sided odd-share control

Hypothesis $\mathrm{H}_q(C, A)$. For all sufficiently large y, every $1 \leq t < d(y)$ and every $L(y)$-bad word $w \in \{O, E\}^t$,

$$\#\{n \in [w]_y :\ \mathrm{word}_{t+1}(n) = wO\} \ \leq\ q\,\#[w]_y + \frac{y}{(\log y)^A}.$$

(The depth-0 cylinder is all odd starts, whose next letter is O with share 1; the hypothesis starts at depth 1. As in Section 8.3, no condition is imposed after an envelope crossing.)

Theorem 9.1 (one-sided form). Let $0 < q < \log 2/\log 3 = 0.6309$, $\mu = 1 - q\log_2 3 > 0$, $C > 1/\mu$, and $e_q(C) = 2(C\mu - 1)^2/(C(\log_2 3)^2 \ln 2)$. Under $\mathrm{H}_q(C, A)$ with $A > C + e_q(C)$, for every $\varepsilon > 0$ and all large y,

$$\#\{n \text{ odd} \in (y, 2y] : \ n \notin R\} \ \le \ \frac{y}{2}\Big(\frac{\log 2y}{\log N_0}\Big)^{-(e_q(C)-\varepsilon)}.$$

Hence $e_q(C) > 1 - \lambda^{**}$ implies the conjecture; the least C is 19 at $q = \frac{1}{2}$, 41 at 0.55, 223 at 0.60, 1586 at 0.62 (with λ^{***}: 18, 39, 206, 1451).

Proof. Let n be uniform on the $N_y = y/2 + O(1)$ odd integers of $(y, 2y]$. Bounded endpoint factors are absorbed using ε-slack. Follow its actual word until the first prefix that is not $L(y)$-bad, and after that crossing append synthetic letters E. Let $\widehat{\mathcal{F}}_t$ be the filtration of these stopped words, $\widehat{X}_t$ the indicator that the synthetic next letter is O, and $\widehat{u}_t = (\log_2 3 - 1) + \sum_{1\le s<t}(\widehat{X}_s \log_2 3 - 1)$. On a bad atom the stopped word is the actual word, so H_q applies; on a crossed atom $\widehat{X}_t = 0$. Thus, with $\eta_t = (\mathbb{P}(\widehat{X}_t = 1 \mid \widehat{\mathcal{F}}_t) - q)^+$, each bad atom $[w]$ has $\eta_t \le y(\log y)^{-A}/\#[w]$, while every crossed atom has $\eta_t = 0$. There are at most 2^t atoms, so $\mathbb{E}[\eta_t] \ll 2^t(\log y)^{-A}$ and $\mathbb{E}[\sum_{1\le t<d}\eta_t] \ll (\log 2y/\log N_0)^C(\log y)^{-A}$. Since $\mathbb{E}[\widehat{u}_{t+1} - \widehat{u}_t \mid \widehat{\mathcal{F}}_t] \le -\mu + \eta_t \log_2 3$, the process $M_t = \widehat{u}_t - \widehat{u}_1 - \sum_{1\le s<t}\mathbb{E}[\widehat{u}_{s+1} - \widehat{u}_s \mid \widehat{\mathcal{F}}_s]$ is a martingale with increments in an interval of length $\log_2 3$, and $\widehat{u}_d \le \widehat{u}_1 + M_d - (d-1)\mu + \log_2 3\sum_{s<d}\eta_s$. Fix $\kappa \in (0, 1)$. If $n \notin R$, the contrapositive of Lemma 8.1 gives $u_t > -L$ for every $t \le d$. Thus no crossing occurred, $\widehat{u}_d = u_d > -L$, and hence either $\log_2 3\sum_s \eta_s > \kappa(d-1)\mu$, an event of probability $O((\log y)^{C-A})$ by Markov's inequality, which is $o((\log y)^{-e_q(C)+\varepsilon})$ under the stated condition, or $M_d > a := (1-\kappa)(d-1)\mu - L - \widehat{u}_1 \ge L((1-\kappa)C\mu - 1) - \log_2 3$. By the Azuma–Hoeffding inequality, $\mathbb{P}(M_d > a) \le \exp(-2a^2/((d-1)(\log_2 3)^2)) \le 2^{-L(e_q^{(\kappa)}(C)-o(1))}$ with $e_q^{(\kappa)}(C) = 2((1-\kappa)C\mu - 1)^2/(C(\log_2 3)^2 \ln 2) \to e_q(C)$ as $\kappa \to 0$. $\square$

So no lower bound on odd shares and no vanishing error are needed: if no $L(y)$-bad cylinder of depth below $41\log_2(\log 2y/\log N_0)$ sends more than 55% of its members to an odd state, every positive integer reaches 1.

9.2 The pressure form

Let $N = y/2$, let $\tau(n) = \min\{t \ge 1 : J^t(n) \le N_0\}$, with $\min\emptyset = \infty$, be the entrance time into the floor, call n *live at depth* t if $\tau(n) > t$, and write $o_t(n)$ for the number of odd letters among $n, J(n), \dots, J^{t-1}(n)$. By Lemma 8.1, $\tau(n) > t$ implies $u_t(n) > -L(y)$. The stopping is essential: $J(1) = 1$ is odd, so an orbit that has reached 1 has an all-O tail and the unstopped odd count of every start in R grows linearly. For $\theta > 0$ put $a_\theta = \frac{1}{2}(1 + e^\theta)$.

Hypothesis $\mathrm{P}_\theta(C)$ (live pressure). For all sufficiently large y, with $d = d(y)$,

$$\frac{1}{N}\sum_{\substack{n\ \text{odd}\in(y,2y] \\ \tau(n)>d}} e^{\theta\, o_d(n)} \ \le \ a_\theta^d\, e^{o(d)}.$$

Theorem 9.2 (pressure form). Assume $\mathrm{P}_\theta(C)$ with $C \ge 5$ and $\theta = \theta_C = \log\big(p_C/(1 - p_C)\big)$. Then for every $\varepsilon > 0$ and all large y,

$$\#\{n \text{ odd} \in (y, 2y] : \ \tau(n) > d(y)\} \ \le \ \frac{y}{2}\Big(\frac{\log 2y}{\log N_0}\Big)^{-(e(C)-\varepsilon)},$$

the bound of Theorem 8.3. Hence $\mathrm{P}_{\theta_C}(C)$ with $C \ge 19$ implies the conjecture using Theorem 1, and with $C \ge 18$ under the conditional exponent of Appendix C.

Proof. If $\tau(n) > d$ then $u_d > -L$, i.e. $o_d \ge p_C d$. Hence

$$\#\{\tau > d\} \le \#\{\tau > d,\ o_d \ge p_C d\} \le e^{-\theta p_C d}\sum_{\tau(n)>d} e^{\theta o_d(n)} \le N\, e^{-\theta p_C d} a_\theta^d e^{o(d)}.$$

At $\theta = \theta_C$, $\theta p_C - \log a_\theta = D(p_C\|\frac{1}{2})$, so the count is $\le N\exp(-dD(p_C\|\frac{1}{2})(1-o(1))) \le N2^{-(e(C)-\varepsilon)L}$. $\square$

Lean: exact form, in `formal/Problems/Juggler/FatePressure.lean`, on the live weight of `LiveCountWeight` (starts in $\{1,\dots,N\}$ that stay above N_0 for d steps). `livePressure` is $\sum_{n \text{ live}} x^{o_d(n)}$ as the generating function of the live weight; `live_count_le_pressure` is the Markov step $\#\{\text{live}, o_d \ge k\} \le \text{pressure}/x^k$; `live_oddCount_ge` is Lemma 8.1 on live starts ($o_d \ge p_C d$ for $d \ge CL(N)$); and `live_count_le_of_pressure` says: if the pressure at the tilt $x = p_C/(1-p_C)$ is at most $Na_\theta^d E$, the live starts number at most $N\exp(-dD(p_C\|\frac{1}{2}))E$, by the identity $a_\theta^d/x^{p_C d} = e^{-dD(p_C\|1/2)}$ (`tilt_value`). The substitution $E = e^{o(d)}$ and $d = \lceil CL\rceil$ that gives the displayed bound is not formalized.

For $1 \le t < d$ let $\mu_{\theta,t}$ be the probability measure on the starts live at depth t with density proportional to $e^{\theta o_t(n)}$, and let $s_\theta(t) = \mu_{\theta,t}(J^t(n) \text{ odd})$ be the *tilted odd share*: the probability that the next letter is O when odd-heavy live prefixes are up-weighted exponentially. If the live set is empty, define $s_\theta(t) = 0$; subsequent live moments are then zero.

Hypothesis $\mathrm{M}_{\theta,q}(C)$ **(no momentum).** For all large y, $\sum_{t=1}^{d(y)-1}\big(s_\theta(t) - q\big)^+ = o(d(y))$.

Proposition 9.3. $\mathrm{M}_{\theta,1/2}(C)$ implies $\mathrm{P}_\theta(C)$. More generally, for $0 < q < 1$ and a fixed $\theta > 0$, $\mathrm{M}_{\theta,q}(C)$ gives

$$\#\{\tau > d\} \le N\exp\{-d[\theta p_C - \log(1-q+qe^\theta) - o(1)]\}.$$

If $0 < q < p_C < 1$ and the hypothesis is assumed at the optimizing tilt $\theta = \theta_{C,q} := \log\frac{p_C(1-q)}{q(1-p_C)}$, this becomes $N\exp(-dD(p_C\|q)(1-o(1)))$, with exponent $e_q^{\mathrm{Ch}}(C) = C\,D(p_C\|q)/\ln 2$, at least the Azuma exponent of Theorem 9.1. Its least C at $q = 0.5, 0.55, 0.60, 0.62$ is $19, 41, 214, 1496$ (with λ^{***}: $18, 38, 198, 1369$).

Proof. Dropping the condition $\tau > t+1$ in favour of $\tau > t$ only enlarges the sum, so

$$\sum_{\tau > t+1} e^{\theta o_{t+1}} \le \sum_{\tau > t} e^{\theta o_{t+1}} = \Big(\sum_{\tau>t} e^{\theta o_t}\Big)\big(1 + (e^\theta - 1)s_\theta(t)\big).$$

With $a_{\theta,q} = 1 + (e^\theta - 1)q$ and $c_\theta = (e^\theta - 1)/a_{\theta,q}$, $1 + (e^\theta-1)s \le a_{\theta,q}\exp\big(c_\theta(s-q)^+\big)$. Telescoping from $t = 1$ gives $\sum_{\tau>d} e^{\theta o_d} \le Ne^\theta a_{\theta,q}^{d-1}\exp\big(c_\theta\sum_{t<d}(s_\theta(t)-q)^+\big)$, which is P_θ at $q = \frac{1}{2}$. For general q, the exponential Markov step gives $\theta p_C - \log a_{\theta,q}$ at the same fixed θ. Assuming $\mathrm{M}_{\theta,q}$ at $\theta = \log\frac{p_C(1-q)}{q(1-p_C)}$ gives $D(p_C\|q)$. $\square$

Lean: `weightGen_succ_le_share` (the one-depth step), `one_add_le_exp_excess`, `weightGen_le_pressure` (the telescoped bound) and `count_le_pressure` (the exponential Markov step), in `formal/Problems/Juggler/TiltedShare.lean`, on the word-weight framework of `RateFreeDensity`; `tilt_exponent_eq_kl` is the identity at the re-centring tilt.

9.3 Scope of the moment criterion

(a) Any $o(\log\log y)$ **initial depths are free.** A letter contributes a factor at most e^θ to the moment, so bias — however extreme — at any set of depths of size $o(d(y))$ is absorbed by the $e^{o(d)}$. In particular the depth-five split (Paper B's Conjecture 7.3) and every split to any fixed depth, or to depth $s_0(y)$ for any $s_0 = o(\log\log y)$, is irrelevant to the reduction.

(b) A tower comparison in a model population. Suppose a model cylinder O^t has size $N\beta^{t-1}$. Its moment relative to the fair *reference* total $Ne^\theta a_\theta^{t-1}$ is $(\beta e^\theta/a_\theta)^{t-1}$. This ratio decays precisely when $\beta < (1+e^{-\theta})/2$, about 0.836 at $\theta_{19} \approx 0.396$. Its count alone exceeds the reference survival exponent only when $\beta > 2^{-e(C)/C}$, about 0.981 at $C = 19$.

These are algebraic comparisons to a reference mass. They do not bound $s_\theta(t)$ in an actual stopped population, whose total live mass may be much smaller. A model tower below 0.836 is therefore not, by itself, a proof of the no-momentum hypothesis.

(c) One-sidedness and aggregation. $\mathrm{M}_{\theta,q}$ bounds the accumulated positive excess of a weighted average over depths. Under-populated odd continuations can offset excess within the same depth; over-populated cylinders contribute in proportion to their tilted weight. In Walsh terms, $e^{\theta X_s} = a_\theta + b_\theta(-1)^{J^s(n)}$ with $b_\theta/a_\theta = -\tanh(\theta/2)$, so the unstopped moment is

$$\frac{1}{N}\sum_n e^{\theta o_d(n)} = a_\theta^d\Big(1 + \sum_{\emptyset\neq T}(-\tanh\tfrac{\theta}{2})^{|T|}\frac{W_T}{N}\Big), \qquad W_T = \sum_n \prod_{s\in T}(-1)^{J^s(n)}:$$

Walsh sums of high order are exponentially down-weighted ($\tanh(0.198) = 0.196$ per letter), whereas $\mathrm{H}(C, A)$ asks for a separate supremum bound on each $L(y)$-bad cylinder at the target depth.

(d) Other sufficient statistics. The proof of Theorem 9.1 also works if the odd-share bound fails only on bad atoms of total probability $O((\log y)^{-B})$ at each depth, with $B > e_q(C)$: the expected accumulated excess is $O(d(\log y)^{-B})$, and Markov's inequality absorbs it.

For comparison, let $D(w) = \#[wO] - \frac{1}{2}\#[w]$, $\mathcal{C}_t = \sum_{|w|=t}\#[w]^2$, and let W_T be the Walsh sums on the same full set of odd starts. Parseval gives the exact identities

$$\sum_w D(w)^2 = 2^{-t-2}\sum_{S\subseteq\{0,\ldots,t-1\}}|W_{S\cup\{t\}}|^2 = \tfrac{1}{2}\mathcal{C}_{t+1} - \tfrac{1}{4}\mathcal{C}_t.$$

An additional quantitative bound on these sums can supply an exceptional-atom estimate. No rate-independent equivalence with the restricted bad-cylinder hypothesis is asserted. Full-population fair collision bounds also face the absorbed-cylinder obstruction of Section 8.3.

(e) Bounded-depth statistics and the walk model. Fix k. The probability measure on parity words that is fair to depth k and all-O afterwards satisfies every cylinder-count statement of depth $\leq k$ exactly, and under it a positive proportion of model walks never crosses the barrier. An argument for the Tao-type bound that used only depth-$\leq k$ cylinder statistics of the Juggler map and the walk logic would prove the same bound for that measure. Thus finite-depth statistics plus the envelope walk alone are insufficient. This model example does not rule out a proof using additional arithmetic properties of the Juggler map.

9.4 Where the difficulty lies

$\mathrm{P}_\theta(C)$ says that the exponentially tilted distribution of live parity words of length $d \asymp \log_2\log y$ has the fair pressure $\log a_\theta$: odd-heavy live words — those with about $e^\theta/(1+e^\theta) \approx 60\%$ odd letters, the words the tilt selects — are not over-populated at an exponential rate in d. Typical such words contain odd runs of length ≥ 4, so the nested towers of height ≥ 4 that stop Paper B at depth five sit inside almost every relevant cylinder; but no individual cylinder is required to split fairly, only the tilted average over the $\approx 2^{0.97d}$ typical cylinders at each depth. This is a statement about averaging over cylinders at unbounded depth — a mean over characters, not a supremum — of a kind for which analytic number theory sometimes has tools where pointwise bounds fail. We record the question and do not pursue an analytic approach here.

10. The free term and limits of the reductions

10.1 The free term is the infinite-depth live mass

Recall the free term ψ_F of (6.1). For odd n, $n \in F \iff \lfloor n^{3/2}\rfloor \in F$ (Theorem 6.1), and $n \in F \iff \tau(n) = \infty$ since $[1, N_0] \subseteq R$. Let $\mathbb{P}_x^{\log}$ be the $1/n$-weighted measure on the odd $n \in (\sqrt{x}, x]$.

Proposition 10.1 (normalized live limit). Put

$$q_{OO}(t) = \frac{8}{t}\sum_{\substack{\sqrt{x}<n\leq x\\ \mathrm{word}_2(n)=OO}}\frac{1}{n}, \qquad x = e^t.$$

Whenever the conditioning set is nonempty,

$$\psi_F(t) = q_{OO}(t) \lim_{d\to\infty} \mathbb{P}_x^{\log}(\tau(n) > d \mid \mathrm{word}_2(n) = OO). \tag{10.1}$$

The limit is decreasing. For large t, $q_{OO}(t) = 1 + O(e^{-\delta t})$ for some $\delta > 0$. If a hypothesis in Sections 8–9 gives a live-start bound with rate $r > 0$, then, for every $0 < \varepsilon < r$,

$$\psi_F(t) \ll_\varepsilon \left(\frac{t}{2\log N_0}\right)^{-(r-\varepsilon)}. \tag{10.2}$$

For the one-sided and no-momentum forms, the same proofs bound live starts as well as failures. The rate is the case-dependent rate in Theorem 4.

Proof. At fixed x the conditioning set is finite, and $F = \{n : \tau(n) = \infty\}$. Continuity for decreasing events gives (10.1), including its normalization factor. The depth-two parity estimate for odd starts gives OO-count $y/4 + O(y^{1-\delta'})$ on dyadic blocks. Partial summation, or the same exponential-sum bound on initial subintervals, gives the asserted reciprocal-mass asymptotic for q_{OO}.

Cover $(\sqrt{x}, x]$ by downward dyadic blocks, allowing one partial block at the lower end. On $(y, 2y]$ the reciprocal mass of live starts is at most their count divided by y, hence $O((\log y/\log N_0)^{-(r-\varepsilon)})$. There are $O(t)$ blocks and $\log y \geq t/2 - O(1)$. Multiplication by $8/t$, the definition of ψ_F, proves (10.2). No equality between an unweighted block ratio and its reciprocal-weighted ratio is used. □

This connects the free term with live counts. The decreasing limit does not provide a rate of convergence in depth; the finite-depth estimates remain additional sufficient hypotheses.

10.2 What the upper recursion proves

Equations (6.1) and the fiber upper bound give, for large t,

$$\varphi_F(t) \leq \tfrac{1}{2}\varphi_F(t/2) + \tfrac{1}{3}\varphi_F(3t/4) + \tfrac{1}{4}\psi_F(t) + O(e^{-\delta t}). \tag{10.3}$$

The coefficient is $1/3$ with the proved upper share; the ideal fiber model would replace it by $1/4$. If c denotes either coefficient, the homogeneous exponent $e_c > 0$ is the root of $\frac{1}{2}2^{e_c} + c(4/3)^{e_c} = 1$. Numerically it is 0.3391 for $c = 1/3$, and $1 - \lambda_{\mathrm{ideal}} = 0.5073$ for $c = 1/4$. These values describe the upper comparison, not a proved exact decay rate of a nonempty failure set.

Corollary 10.2 (upper comparison). If $\psi_F(t) = O(t^{-r})$ for some $r > 0$, then $\varphi_F(t) = O(t^{-a})$ for every $0 < a < \min(r, e_c)$, using $c = 1/3$ in (10.3). The ideal model has the analogous assertion with $c = 1/4$. The upper inequality alone does not imply $F = \emptyset$.

Proof. For such a, $h = \frac{1}{2}2^a + c(4/3)^a < 1$. Choose a large initial scale T and then K so that $\varphi_F(t) \leq Kt^{-a}$ on $[T/2, T]$, and so that the forcing in (10.3) is bounded by $(1-h)Kt^{-a}$ for $t \geq T$. Induction on intervals $[T, (4/3)^j T]$ proves the bound. At the endpoint $a = e_c$ there is no slack; in particular an endpoint estimate cannot be inferred by the same argument. A positive multiple of t^{-e_c} satisfies the homogeneous comparison on a tail, so that comparison alone does not force the function to vanish. The lower contagion estimates and this upper comparison do not establish a matching asymptotic. □

10.3 The depth-uniformity budget

Section 9.3(e) limits what fixed-depth statistics plus the envelope walk alone can prove. The form of Theorem 8.3 says how uniform that information must be *along the cylinder-count route.*

Proposition 10.3 (depth-uniformity budget). Suppose the available cylinder discrepancy bound is

$$\left|\#[w]_y - 2^{-(d-1)}y/2\right| \leq K2^{\kappa d} y^{1-\delta_0 2^{-cd}},$$

for the relevant bad words, with constants $K, \delta_0, c > 0$, $\kappa \geq 0$, and a scale threshold independent of d. At $d = C \log_2 \log y + O(1)$, the resulting error majorant after summing over at most 2^d words is $o(y(\log y)^{-r})$ for every fixed $r > 0$ if and only if $cC < 1$.

Proof. Divide the error majorant by $y(\log y)^{-r}$ and take logarithms. The result is

$$O(1) + ((\kappa + 1)C + r) \log \log y - \Theta((\log y)^{1-cC}).$$

It tends to $-\infty$ exactly when $cC < 1$. If $cC \geq 1$, this majorant is insufficient; the actual error may still be much smaller. Without uniform control of the prefactor and the scale threshold, a bound at every fixed depth would not justify any conclusion at growing depth. □

At $C = 19$ this requires the saving exponent to lose less than a factor $2^{1/19} = 1.037$ per depth; at $C = 41$, $2^{1/41} = 1.017$. Weyl differencing loses a factor 2^c with $c \geq 1$ per differencing (Paper B's chain $\frac{1}{24} \to \frac{1}{96}$ from depth three to four is $c = 2$), and with such a loss the cylinder-count route reaches depth $\approx \log_2 \log y$ — one unit of $L(y)$, enough for the all-E word to reach the floor, not for the Chernoff tail, which needs $C \geq 5$ units. The proposition is narrow by design: it concerns methods whose only available error estimate deteriorates exponentially in the depth and are used through the per-cylinder count of Theorem 8.3. A method that organizes its errors differently, that proves a polylogarithmic relative saving uniformly in the depth (as $\mathrm{H}(C, A)$ asks), or that addresses the pressure statement directly is not covered by it.

11. Numerical observations

The following archived numerical experiments accompany the theorems. Orbit steps and integer counts use exact arithmetic; phases, weighted statistics and fair-coin comparisons use floating-point arithmetic. Random samples use fixed seeds (`research.juggler_sequence.fate_contagion`, `research.juggler_sequence.tao_reduction`; artifacts in Appendix B), they do not establish asymptotic estimates and they enter no analytic proof. Detailed tables are kept in the repository notes [14].

Fibers, blocks, closure. Over $3375 \leq m < 10^5$ and spot ranges at 10^6 and 10^7, the mean of G_m/H_m is 0.5000, 0.5004, 0.5027; the minimum on good fibers is 0.328 ($m = 1003635$, $\alpha_m \approx \frac{1}{3}$); no good fiber falls below $\frac{1}{3}H_m - 2$. On the blocks $m' \in [20, 60) \cup [200, 230) \cup [1000, 1010) \cup \{3000, 5000\}$ the deviation of $|U(m')|$ from its main term is at most $0.72\sqrt{|I(m')_{\mathrm{odd}}|}$, i.e. square-root scale.

The share law and the extremes. A complete census of $\Phi(m)$ for $m \leq 2 \cdot 10^5$ (`research.juggler_sequence.oe_fiber_share`) finds 2481 fibers with $G_m = 0$ and 2508 with $G_m = H_m$, and $H_m \geq 1$ throughout. The mean of G_m/H_m is 0.498 on the decade $[10^2, 10^3)$ and 0.500 on every decade above it, and the empty rate falls 8.0, 3.0, 1.35, 0.90 per cent over the decades from 10^2 to $2 \cdot 10^5$, tracking $1/H_m$. The archived sampling code uses $\widehat{\beta}_m = \alpha_m H_m$, whereas Lemma 4.5 uses $\beta_m = \alpha_m(H_m - 1)$. Near 10^6, 432 fibers with $|\widehat{\beta}_m| \leq 1.2$ give mean absolute model error 0.0195 against $1/H_m = 0.015$, and every extreme fiber found has $\widehat{\beta}_m \in [-0.79, 0.15]$, inside the model window of Corollary 4.6(2). The extremes sit $\asymp m^{1/3}$ apart as Remark 4.8 predicts: in $m \in [10^5, 1.2 \cdot 10^5)$, where that period is 46.4, their median spacing is 48 and 244 of 306 gaps are one period. Empty fibers occur singly or in adjacent pairs and never in threes. The observed empty rate exceeds the $(25/108)/H_m$ of Corollary 4.6(3) by about 1.4, and the sample is consistent with a discretization effect: of the 297 empty fibers in $[10^5, 1.3 \cdot 10^5)$, 174 have $S(\widehat{\beta}_m, \theta_m) = 0$ while 123 have $S > 0$ and miss only because the fiber samples it at 31 points. Consistently, the ratio falls 1.49, 1.44, 1.36 across the last three decades. The closure of the Lean-verified seed $[1, 260]$ under the two productions of Section 3, computed exactly up to 10^9, has density between 0.45 and 0.49 on the complete dyadic blocks $(2^k, 2^{k+1}]$, $16 \leq k \leq 28$. The last reported block is truncated to $(2^{29}, 10^9]$; its density is 0.4529, against 0.25 for the E-only closure, and its realized recursion coefficients at 10^9 are 0.9998 for E (theory 1) and 0.3332 for OE (reference $\frac{1}{3}$). The separate decomposition sample at $x = 10^6$ has OO-type share 0 against reference share 0.2987. This finite closure is not a two-way closed fate class; its omitted production illustrates the OO term without establishing an asymptotic property of F.

Survival above the certified floor. Because every $n \le N_0 = 3.5 \cdot 10^8$ reaches 1, the statistic "$J^t(n) \le N_0$ for some $t \le d$" is a finite computation, and its complement is the live set of Theorem 8.3 (contained in the envelope-bad cylinders). For random odd starts in $(y, 2y]$, the fraction still above N_0 after d steps is compared with the odd-start fair-coin survival $\Pr[u_t > -L(y)\ \forall t \le d \mid u_1 = \log_2 3 - 1]$:

y	$L(y)$	$d = 11$	$d = 21$	$d = 31$	$d = 36$
10^{12}	0.526	0.221/0.221	0.082/0.084	0.037/0.038	0.027/0.028
10^{15}	0.841	0.251/0.267	0.100/0.106	0.044/0.047	0.032/0.035
10^{20}	1.249	0.295/0.309	0.135/0.138	0.068/0.068	0.049/0.049
10^{30}	1.826	0.442/0.439	0.175/0.175	0.098/0.095	0.075/0.070
10^{50}	2.558	0.554/0.554	0.267/0.264	0.148/0.143	0.101/0.097

Entries are empirical/fair at the depths retained in the archived record: 40000 starts at $10^{12}, 10^{15}, 10^{20}$, and 20000 at $10^{30}, 10^{50}$. The sampler draws integers in $(y, 2y]$ and rounds even draws up to the next odd integer; this differs from uniform odd sampling only at the two endpoints, with total variation $1/y$. A resource cap is conservatively recorded as no observed descent, so empirical survival is an upper bound for the sampled live fraction when a cap is reached.

The separate pressure runs use 40000 starts at each of $10^{20}, 10^{30}, 10^{50}$, with no capped orbits. At these scales, the recorded extrema of $s_\theta(t)$ over $1 \le t \le 40$ and $\theta \in \{0.396, 0.6\}$ are approximately 0.4728 and 0.5337. At the four recorded depths $10, 20, 30, 40$, the live-moment ratio to the fair-coin comparison ranges from 0.9521 to 1.0812, using outward rounding. The archived 10^{12} pressure run has one capped orbit; its unknown later letters are omitted by that estimator. It is excluded from this pressure comparison and cannot certify an upper bound on the full live moment. These finite observations do not establish the hypotheses of Section 9 at depths tending to infinity.

12. Conclusions and open estimates

The main unconditional result is the logarithmic counting lower bound for every nonempty backward-closed set. The resulting eventual-entry equivalence requires no conjectural cylinder estimate. Theorems 8.3, 9.1 and 9.2 are implications from explicit unproved hypotheses; Appendix C is conditional on an additional localized discrepancy estimate. Selected finite computations and Lean proofs have the scopes listed in Section 1.4 and Appendix A.

One sufficient remaining target is

$$\#\{n \text{ odd in } (y, 2y] : \tau(n) > \lceil CL(y) \rceil\} \ll y(\log y)^{-e}, \qquad e > 1 - \lambda^{**},$$

for a fixed C and all sufficiently large y. This would imply universal termination by Theorem 7.2. Universal termination alone gives no uniform time bound of this form in the present argument. The equivalent statement in Theorem 7.3 concerns $\tau(n) = \infty$, not $\tau(n) > \lceil CL(y) \rceil$.

Live pressure at the optimizing tilt is one sufficient route to this target. The no-momentum hypothesis is a sufficient condition for that pressure bound, not a proved equivalent condition. The ideal production calculations describe possible exponents under additional fiber assumptions. They neither prove those assumptions nor classify every possible route to termination. No nontrivial cycle or unbounded orbit is excluded by this paper.

Appendix A. Lean names

All in `formal/Problems/Juggler/FateContagion.lean` unless noted. The root `formal/Problems/JugglerFatePaper.lean` imports exactly the twelve modules named here and builds with `lake build Problems.JugglerFatePaper` without `sorry` and without `native_decide`; `formal/AxiomCheckPaperC.expected` records the axioms of every name below. Lean certifies the exact combinatorial identities and the two abstract lemmas listed here, not the analytic density estimates.

Statement	Lean
backward closure	`BackwardClosed`, `backwardClosed_iterate`
Lemma 2.1, fate classes closed	`reachesOne_backwardClosed`, `not_reachesOne_backwardClosed`, `ancestor_backwardClosed`, `escapes_backwardClosed`, `reachesOne_floorPower`, `escapes_floorPower`
Lemma 2.1, trichotomy	`Periodic`, `periodic_of_repeat`, `fate_trichotomy` (from `cycles_or_escapes`)
Lemma 2.1, exclusion	`reachesOne_bounded`, `reachesOne_not_escapes`, `cycle_basin_bounded`, `cycle_basin_not_escapes`, `reachesOne_not_cycle_basin`, `periodic_iterate_mod`
Lemma 3.1 (even block)	`floorPower_even_block`, `even_block_mem`, `even_block_card`
Lemma 3.2 (OE fiber)	`sqrt_sqrt_eq_iff`, `floorPower_oe_fiber`, `oe_fiber_mem`, `oe_fiber_disjoint`
Theorem 6.1 (odd generation)	`ForwardClosed`, `reachesOne_forwardClosed`, `not_reachesOne_forwardClosed`, `escapes_forwardClosed`, `mem_iff_floorPower_mem`, `exists_odd_ancestor`, `exists_odd_ancestor_ge_three`, `nonempty_iff_odd_image_mem`, `odd_mem_iff`
Lemma 8.1 (envelope descent)	`iterate_le_of_envelope`, `mem_of_envelope_floor`, `reachesOne_of_itinerary_envelope`; power envelope `power_bound_word` (Paper A layer)
Lean floor $N_0 = 260$	`reachesOne_of_lt_two_hundred_sixty_one`
Lemma 4.7 (cube fibers), in `Problems/Juggler/CubeFiber.lean`	`cube_fiber_range`, `cube_fiber_sqrt_even`, `cube_fiber_even_image`, `even_cube_fiber_full`, `cube_fiber_sqrt_odd`, `cube_fiber_alternating`, `odd_cube_fiber_alternating`
Lemma 4.1 (sweep), in `Problems/Juggler/FateSweep.lean`	`Sweep.cell`, `Sweep.sweep_cell`, `sweep_fract_lt_half`, `sweep_fract_ge_half`, `sweep_ceil`, `sweep_rep_le_half`, `sweep_rep_gt_half`
Lemma 4.1' (monotone pairing), in `Problems/Juggler/FateSweepMonotone.lean`	`Sweep.sweep_monotone_cell`, `sweep_monotone_fract_lt_half`, `sweep_monotone_fract_ge_half`, `sweep_monotone_ceil`, `sweep_monotone_rep_le_half`, `sweep_monotone_rep_gt_half`

Statement	Lean
Lemma 5.1 (recursion), in `Problems/Juggler/FateRecursion.lean`	`recursion_lemma`
Section 6.2, Proposition 6.3(i), in `Problems/Juggler/FateFirstLetter.lean`	`MinimalMember`, `minimalMember_odd`, `minimalMember_image_odd`, `minimal_failure_odd_odd`, `exists_minimal_failure`, `first_letter_trichotomy`, `first_letter_pieces_disjoint`
Lemma 8.2 (Chernoff count), in `Problems/Juggler/FateChernoff.lean`	`weightGen_one`, `count_oddCount_ge_le`, `count_oddCount_ge_real_le`, `entropyLog`, `klHalf`, `klHalf_eq`, `klHalf_nonneg`, `tilt_value`, `count_oddCount_ge_le_exp`, `count_oddCount_ge_le_kl`, `LBad`, `pC`, `chernoffExponent`, `logb_two_three_le`, `half_le_pC`, `pC_lt_one`, `LBad_oddCount_ge`, `LBad_count_le`
Theorem 8.3 (explicit form), in `Problems/Juggler/FateChernoff.lean`	`cylinder`, `oddFailures`, `EnvelopeBad`, `oddFailures_subset_bad_cylinders`, `oddFailures_card_le`, `LBad_of_envelopeBad`, `oddFailures_card_le_chernoff`, `scaleRatio`, `scaleL`, `depth`, `cylinder_even_root_empty`, `oddFailures_card_le_explicit`
Theorem 9.2 (pressure form), in `Problems/Juggler/FatePressure.lean`	`livePressure`, `live_count_le_pressure`, `envelopeBad_of_liveTo`, `LBad_of_liveTo`, `live_oddCount_ge`, `live_count_le_of_pressure`
Proposition 9.3 (pressure telescoping), in `Problems/Juggler/TiltedShare.lean`	`oddMass`, `tiltedShare`, `weightGen_succ_le_share`, `one_add_le_exp_excess`, `weightGen_le_pressure`, `count_le_pressure`, `NoMomentum`, `count_le_of_noMomentum`, `tilt_exponent_eq_kl`, `MeanShare`, `weightGen_le_of_meanShare`, `MeanShareOff`, `initial_depths_are_free`, `tower_ratio_lt_one`
Lemma 5.2 (seed), in `Problems/Juggler/FateSeed.lean`	`exists_ge_three_of_backwardClosed`, `seed_lemma`, `seed_constant_pos`
Theorem 7.2 (Tao-type rate, contagion as a hypothesis), in `Problems/Juggler/FateTaoReduction.lean`	`logMass_le_oddLogMass`, `oddLogMass_le_of_dyadic`, `tao_rate_implies_empty`, `tao_rate_implies_conjecture`
Lemmas 4.2–4.3, Proposition 4.4, Theorem 5.3, Theorem 7.3, Sections 8–10 except Lemma 8.2, the explicit form of Theorem 8.3, the exact form of Theorem 9.2 and Proposition 9.3, Appendix C	human proofs

Appendix B. Constants and artifacts

Exponents. Roots of $2^{-\lambda} + \sum_i c_i e_i^{\lambda} = 1$:

recursion	terms (c_i, e_i)	root
elementary sweep only	$(\frac{2}{21}, \frac{3}{4})$	0.1385
block average only (λ^*)	$(\frac{1}{3}, \frac{3}{8})$	0.3774
block average + sweep (adversarial 1/7)	$(\frac{5}{21}, \frac{3}{8}), (\frac{2}{21}, \frac{3}{4})$	0.4051

recursion	terms (c_i, e_i)	root
block average + pairing (named intermediate)	$(\frac{1}{9}, \frac{3}{8}), (\frac{2}{9}, \frac{3}{4})$	0.4480
+ $OEOEE$ (named intermediate; elementary, audited)	$\ldots, (\frac{1}{27}, \frac{9}{32})$	0.4801
+ $V_3 = OEOEOEE$ (named intermediate; elementary, audited)	$\ldots, (\frac{1}{81}, \frac{27}{128})$	0.4891
+ $V_4 = OEOEOEOEE$ (named intermediate; elementary, audited)	$\ldots, (\frac{1}{243}, \frac{81}{512})$	0.4916
+ $V_5 = OEOEOEOEOEE$ (named intermediate; elementary, audited)	$\ldots, (\frac{1}{729}, \frac{243}{2048})$	0.4924
+ $V_6 = OEOEOEOEOEOEE$ (λ^{**}, Theorem 1; elementary, audited)	$\ldots, (\frac{1}{2187}, \frac{729}{8192})$	0.4926
pairing + $OOEEE$ (λ^{***}, Appendix C, conditional)	pairing terms and $(\frac{1}{9}, \frac{9}{32})$	0.5392
depth-two ideal-share model	$(\frac{1}{3}, \frac{3}{4})$	0.4927
+ $OOOEE$, $OOEOE$ (closed: fibers $P^{5/32}$; Lemma 3.9 leftover $P^{89/96}$)	$\ldots, (\frac{2}{27}, \frac{27}{64})$	0.5561
pairing + $OEOEE$ + $OOEEE$ (model comparison)	pairing terms and $(\frac{4}{27}, \frac{9}{32})$	0.5665
conditional infinite $V_k = (OE)^{k-1}OEE$ model	$\ldots, (3^{-(k+2)}, (\frac{3}{4})^k \frac{3}{8})_{k\geq 1}$	0.4927
conditional infinite V_k model and $OOEEE$		0.5769
conditional O-runs ≤ 2, pairing share (Section 5.7)	transfer matrix	0.6247
conditional O-runs ≤ 2, ideal-share model	transfer matrix	0.7180
conditional O-runs ≤ 3, pairing share	transfer matrix	0.7095
conditional O-runs ≤ 3, ideal-share model	transfer matrix	0.8414
abstract ceiling, all O-runs at the ideal share (Proposition 5.12)	$2^{-\lambda} + \frac{1}{3}(\frac{3}{2})^{\lambda} = 1$	1

Depth constants. The least integer values below use the three explicitly named rate thresholds; intermediate V_2 through V_5 have complementary thresholds 0.5199, 0.5109, 0.5084, and 0.5076, respectively.

criterion	λ^{**}, rate 0.5074	pairing, rate 0.5520	λ^{***}, rate 0.4608
fair Chernoff	19	20	18
one-sided Azuma, $q = 0.5, 0.55, 0.60, 0.62$	19, 41, 223, 1586	20, 44, 240, 1715	18, 39, 206, 1451
optimized pressure, same q	19, 41, 214, 1496	20, 43, 230, 1618	18, 38, 198, 1369

Depth $d(y) = \lceil 20L(y) \rceil$ (the conservative a-fortiori table) with $N_0 = 3.5 \cdot 10^8$: 25 at 10^{20}, 72 at 10^{100}, 138 at 10^{1000}, 204 at 10^{10000}.

Decimal values in the tables are rounded; all strict inequalities use the roots of the displayed defining equations.

Artifacts (repository `sneakyweasel/btlab`; SHA-256):

- `data/research/juggler/fate_contagion/summary.json`
  SHA-256: `85030bcb5f4964b814b101683c2721efa5f7299b687afa9e399febe60343a10c`
- `data/research/juggler/tao_reduction/summary.json`
  SHA-256: `76c0ae713d34569cdf8efd90231712f9281f7d5083346a2d9384c536f7cd34cc`
- `formal/Problems/Juggler/FateContagion.lean`
  SHA-256: `cac6a00884346fcc98a03603bf919e03a681f8f66b8b55f3cff9d336e83a2472`
- `formal/Problems/Juggler/CubeFiber.lean`
  SHA-256: `acbf621b4651782eff7922512d096832aa20ea59a59261839b3e5926189dedba`
- `formal/Problems/Juggler/TiltedShare.lean`
  SHA-256: `cfb1b2b6cbe08be3a9acc1f4fa33c9afaa2dfa73beed2125dbe0793c540e239e`
- `formal/Problems/Juggler/FateRecursion.lean`
  SHA-256: `13de9eb27d4e34b58d783d589574838028d767bcb5ea3f2bb8449e74e3be04d3`
- `formal/Problems/Juggler/FateFirstLetter.lean`
  SHA-256: `635bb6163f5a054085087eefcd2c62037c61538fe863d53e9dca69da8fddb3e9`
- `formal/Problems/Juggler/FateSweep.lean`
  SHA-256: `71c5c2472d2c2d31d7b2565e66f92e97b3ae9bd0c76aa45a1703eea7bd910d39`
- `formal/Problems/Juggler/FateChernoff.lean`
  SHA-256: `4eec5c05916226374815b586252aa8c7a147615d27088939044f80db3b94f515`
- `formal/Problems/Juggler/FatePressure.lean`
  SHA-256: `f69ad74fcaed87b692451113cf72eeadc1182efc0ac8121e255e6b079bf2bbf2`
- `formal/Problems/JugglerFatePaper.lean`
  SHA-256: `cb096d601d6c5ef6d19e7c7b81a52ae8a708f241d8311eb89794a699cec3af2b`
- `formal/AxiomCheckPaperC.expected`
  SHA-256: `5939a49ce68518bbff9544fc34287bb41fdab54c11d2a2ba7817dfc2ee99f523`
- `src/research/juggler_sequence/fate_contagion.py`
  SHA-256: `84bafabf891e9be8c764eed222d7b72a96b4971708502cbb7dd43287deaf6ec0`
- `src/research/juggler_sequence/tao_reduction.py`
  SHA-256: `73e6cfb53a094ccbc998eff6bbf422361fc6a7cbef5f32807215bae6a41c0471`
- `docs/theory/figures/render_paper_c_figures.py`
  SHA-256: `fa186851c24f9c4bbf78ffaff6eff1e90627019c4737a99250087ed769d6dc82`

The archived records retain the parameter names and values of their original runs (including the older pairing-only threshold). Current exponents and depth constants are checked separately in `data/research/juggler/paper_c_publication_review/validation.json`. The numerical experiments of Section 11 use fixed random seeds. Their code is run by `python -m research.juggler_sequence.fate_contagion` and `python -m research.juggler_sequence.tao_reduction`. The current code reports current rate constants; archived parameter labels are preserved in the supplied JSON. The figures are regenerated by `python docs/theory/figures/render_paper_c_figures.py`.

Appendix C. A conditional strengthening

This appendix depends on one analytic hypothesis beyond the main text. No statement of Sections 1–12 uses it; it improves the constants of Theorems 1, 3 and 4 and nothing else.

C.1 The hypothesis

For odd n put $m = \lfloor n^{3/2} \rfloor$, $v = \lfloor m^{3/2} \rfloor$, $\psi(y) = (-1)^{\lfloor y \rfloor}$, and

$$\psi_1 = \psi(n^{3/2}), \qquad \psi_2 = \psi(m^{3/2}), \qquad \psi_3 = \psi(v^{1/2}),$$

the signs of the parities of m, of $\lfloor m^{3/2} \rfloor$ and of $\lfloor v^{1/2} \rfloor$ — the second, third and fourth letters of the itinerary of n along the branch OOE.

Hypothesis L (localized triple parity discrepancy with a slow twist). For every $\varepsilon > 0$ there is P_0 such that for all $P \geq P_0$, every interval $I \subseteq (P, 2P]$ of length $Y \geq P^{1/2}$, every $(a, b, c) \in \{0, 1\}^3 \setminus \{(0, 0, 0)\}$, and every integer ℓ with $|\ell| \leq P^{1/24}$,

$$\Big| \sum_{n \in I,\ n \text{ odd}} \psi_1^a \psi_2^b \psi_3^c \, e\big(\tfrac{\ell}{2} n^{9/16}\big) \Big| \leq Y P^{-1/24+\varepsilon}.$$

Status. Hypothesis L is Theorem 4.12 of the working draft [12] (Section 3.5 there), whose Lemma 4.10 removes the twist by partial summation after the Weyl differencing and whose proof runs the seven-step argument of its Theorems 4.4 and 4.7 (exact linearization of the nested floors, differencing with $H = P^{1/12}$, a cell decomposition, second-derivative tests per cell) over a sub-dyadic interval, exhibiting the three absolute terms — two partial end cells $5.2P^{3/8}$, the pure passenger $3.8P^{7/16}$, the majorant end cells $17P^{1/4}$ — that do not scale with the number of summands. Because [12] is an unrefereed working draft, we keep the statement as a hypothesis here and rely on nothing but the statement; what *is* proved in this appendix is everything that follows from it. (The derivation below coincides with Corollary 4.13 of [12].)

C.2 The $OOEEE$ production on even blocks

For $m' \geq 2$ let $I(m') = [m'^{32/9}, (m'+1)^{32/9})$, an interval of length $Y = \frac{32}{9} m'^{23/9}(1 + O(1/m'))$ at scale $P = m'^{32/9}$, so that $Y \asymp P^{23/32} \geq P^{1/2}$. Let

$$\mathcal{O}(m') = \{n \text{ odd} \in I(m') : \ \text{word}_5(n) = OOEEE,\ J^5(n) = m'\}.$$

Lemma C.1 (transparent nesting at the fifth letter). For odd $n \geq 3$, with m, v as above, $0 \leq n^{9/16} - v^{1/4} \leq n^{-15/16}$. Consequently $\lfloor v^{1/4} \rfloor = \lfloor n^{9/16} \rfloor$ unless $\{n^{9/16}\} < n^{-15/16}$.

Proof. $v \leq m^{3/2} \leq n^{9/4}$ gives $v^{1/4} \leq n^{9/16}$. For the other direction, $m \geq n^{3/2} - 1$ and $v \geq m^{3/2} - 1$ give $v \geq (n^{3/2}-1)^{3/2} - 1 \geq n^{9/4} - \frac{3}{2}n^{3/4} - 1$, and then, by the mean value theorem for $u \mapsto u^{1/4}$ on $[v, n^{9/4}]$, $n^{9/16} - v^{1/4} \leq (n^{9/4} - v)/(4v^{3/4}) \leq (\frac{3}{2}n^{3/4} + 1)/(4v^{3/4})$. Since $v \geq \frac{1}{2}n^{9/4}$ for $n \geq 3$, $v^{3/4} \geq 2^{-3/4}n^{27/16}$, and the bound is at most $2^{3/4}(\frac{3}{2}n^{3/4} + 1)/(4n^{27/16}) \leq n^{-15/16}$ for

$n \geq 3$. The floors of two numbers in $[n^{9/16} - n^{-15/16}, n^{9/16}]$ differ only if an integer lies in that interval, i.e. if $\{n^{9/16}\} < n^{-15/16}$. $\square$

Lemma C.2 (the exceptional set is small). Let $\mathcal{E}(I) = \{n \text{ odd} \in I : \ \{n^{9/16}\} < n^{-15/16}\}$. For $I \subseteq (P, 2P]$ of length $Y \geq P^{1/2}$, $|\mathcal{E}(I)| \ll YP^{-1/16}$.

Proof. $\mathcal{E}(I) \subseteq \{n \text{ odd} \in I : \ \{n^{9/16}\} \in [0, P^{-15/16})\}$. By the Erdős–Turán inequality [15, Theorem 2.5], for any $H \geq 1$,

$$|\mathcal{E}(I)| \ \leq \ \tfrac{Y}{2}P^{-15/16} + C\Big(\frac{Y}{H} + \sum_{h=1}^{H} \frac{1}{h}\Big| \sum_{n \in I \text{ odd}} e(hn^{9/16})\Big|\Big).$$

Write $n = 2k+1$ and $F_h(k) = h(2k+1)^{9/16}$; then $F_h'(k) = \frac{9}{8}h(2k+1)^{-7/16}$ is monotone with $0 < F_h' \leq \frac{9}{8}hP^{-7/16} \leq \frac{1}{2}$ for $h \leq H := \frac{4}{9}P^{7/16}$, so the Kusmin–Landau inequality [13, Theorem 2.1] gives $|\sum_k e(F_h(k))| \ll 1/\min F_h' \ll P^{7/16}/h$. Hence $|\mathcal{E}(I)| \ll YP^{-15/16} + YP^{-7/16} + P^{7/16}$, and $P^{7/16} \leq YP^{-1/16}$ for $Y \geq P^{1/2}$. $\square$

Lemma C.3 (a Fourier expansion of the fifth letter). Let $J = \lfloor P^{1/24} \rfloor$. There are coefficients b_q, $0 < |q| \leq J$, with $|b_q| \ll 1/|q|$, and a nonnegative trigonometric polynomial Δ_J of degree J with all coefficients at most $1/(J+1)$, such that for every real y, $\big|\psi(y) - \sum_{0<|q|\leq J} b_q e(qy/2)\big| \leq 2\Delta_J(y/2)$. Moreover, for $I \subseteq (P, 2P]$ of length $Y \geq P^{1/2}$,

$$\sum_{n \in I \text{ odd}} \Delta_J\big(\tfrac{1}{2}n^{9/16}\big) \ll YP^{-1/24} + P^{7/16}\log P, \qquad \Big|\sum_{n \in I \text{ odd}} e\big(\tfrac{q}{2}n^{9/16}\big)\Big| \ll \frac{P^{7/16}}{|q|} \quad (0 < |q| \leq J).$$

Proof. $\psi(y) = 2 \cdot \mathbf{1}[\{y/2\} < \frac{1}{2}] - 1$, and Vaaler's theorem [10] approximates the indicator of $[0, \frac{1}{2})$ modulo 1 by a trigonometric polynomial of degree J with coefficients $\ll 1/|q|$ and constant term $\frac{1}{2}$, with error at most a majorant Δ_J of the stated form; the constant terms cancel in $2 \cdot \mathbf{1} - 1$. For the two sums, write $n = 2k+1$: the phase $\frac{q}{2}(2k+1)^{9/16}$ has derivative $\frac{9q}{16}(2k+1)^{-7/16}$, monotone, of absolute value between $\frac{9}{16}|q|(2P)^{-7/16}$ and $\frac{9}{16}P^{1/24-7/16} < \frac{1}{2}$, so Kusmin–Landau gives $\ll P^{7/16}/|q|$. The Δ_J sum is its constant term, $\leq Y/(J+1)$, plus $(J+1)^{-1}\sum_{0<|r|\leq J} |\sum_n e(\frac{r}{2}n^{9/16})| \ll P^{7/16}\log J/J$. $\square$

Proposition C.4 (conditional on Hypothesis L). For $m' \geq 2$,

$$|\mathcal{O}(m')| = \tfrac{1}{16}\#\{n \text{ odd} \in I(m')\} + O_\varepsilon\big(|I(m')|\, m'^{-4/27+\varepsilon}\big),$$

every $n \in \mathcal{O}(m')$ has $J^4(n) \in E(m')$, and for a backward-closed A and $m' \in A$, $\mathcal{O}(m') \subseteq A$ with log-mass at least $\frac{1}{9m'}\big(1 - O(m'^{-4/27+\varepsilon})\big)$.

Proof. Along $OOEEE$ the states $J^0(n), \ldots, J^4(n)$ are n (odd), m (odd), v (even), $\lfloor v^{1/2} \rfloor$ (even), $\lfloor \sqrt{\lfloor v^{1/2} \rfloor} \rfloor = \lfloor v^{1/4} \rfloor$ (even; the inner identity is the exact nesting of square roots, Lemma 3.2), and $J^5(n) = \lfloor \sqrt{J^4(n)} \rfloor$. If $J^5(n) = m'$ then $J^4(n) \in [m'^2, (m'+1)^2)$ is even, i.e. $J^4(n) \in E(m')$, so $n \in A$ by Lemma 3.1 and backward closure. For the count, let $I = I(m')$, $P = m'^{32/9}$, $Y = |I|$. For $n \in I \setminus \mathcal{E}(I)$, Lemma C.1 gives $\lfloor v^{1/4} \rfloor = \lfloor n^{9/16} \rfloor$, and $n \in I$ means $n^{9/16} \in [m'^2, (m'+1)^2)$; hence for such n with $\mathrm{word}_5(n) = OOEEE$, $J^4(n) = \lfloor n^{9/16} \rfloor \in [m'^2, (m'+1)^2)$ and $J^5(n) = m'$ automatically. Therefore

$$\big|\, |\mathcal{O}(m')| - \#\{n \text{ odd} \in I : \ \mathrm{word}_5(n) = OOEEE\} \,\big| \leq |\mathcal{E}(I)| \ll YP^{-1/16}$$

by Lemma C.2. Next, for odd n the indicator of $OOEEE$ is $\frac{1}{16}(1-\psi_1)(1+\psi_2)(1+\psi_3)(1+\psi_4)$ with $\psi_4 = \psi(v^{1/4})$: the factor $1 - \psi_1$ enforces m odd, so that $J^2(n) = v$ and ψ_2 is its parity sign; $1 + \psi_2$ enforces v even, so that $J^3(n) = \lfloor v^{1/2} \rfloor$ and ψ_3 is its sign; $1 + \psi_3$ enforces $J^3(n)$ even, so that $J^4(n) = \lfloor v^{1/4} \rfloor$ and ψ_4 is its sign. Expanding the product gives the main term $\frac{1}{16}\#\{n \text{ odd} \in I\}$ and fifteen sign sums $\pm\frac{1}{16}\sum_n \psi_1^a\psi_2^b\psi_3^c\psi_4^d$, $(a,b,c,d) \neq 0$.

Sums with $d = 0$. These are Hypothesis L with $\ell = 0$: each is $\leq YP^{-1/24+\varepsilon}$.

Sums with $d = 1$. Replace $\psi_4 = \psi(v^{1/4})$ by $\psi(n^{9/16})$: by Lemma C.1 the two agree off $\mathcal{E}(I)$, so the change is $\le 2|\mathcal{E}(I)| \ll YP^{-1/16}$. Then expand $\psi(n^{9/16})$ by Lemma C.3:

$$\sum_n \psi_1^a\psi_2^b\psi_3^c\,\psi(n^{9/16}) = \sum_{0<|q|\le J} b_q \sum_n \psi_1^a\psi_2^b\psi_3^c\,e\big(\tfrac{q}{2}n^{9/16}\big) + O\Big(\sum_n \Delta_J\big(\tfrac12 n^{9/16}\big)\Big),$$

using $|\psi_i| = 1$ for the error. If $(a,b,c) \ne 0$, each inner sum is Hypothesis L with $\ell = q$, $|q| \le J \le P^{1/24}$, and $\sum_q |b_q| \ll \log P$, so the total is $\ll YP^{-1/24+\varepsilon}\log P$. If $(a,b,c) = 0$ the inner sums are the pure sums of Lemma C.3, $\ll P^{7/16}/|q|$, and their total against $|b_q| \ll 1/|q|$ is $\ll P^{7/16}$. The majorant term is $\ll YP^{-1/24} + P^{7/16}\log P$ by Lemma C.3.

Collecting, with $P^{7/16}\log P \le YP^{-1/16}\log P$ for $Y \ge P^{1/2}$, every error is $\ll YP^{-1/24+\varepsilon}$, and $P^{-1/24} = m'^{-4/27}$. For the log-mass, each $n \in \mathcal{O}(m')$ has $1/n \ge (m'+1)^{-32/9}$, and $\frac{1}{16}\cdot\frac12\cdot\frac{32}{9}m'^{23/9} = \frac19 m'^{23/9}$. □

C.3 The improved exponent

Theorem C.5 (conditional on Hypothesis L). Theorem 5.3 holds for every $\lambda < \lambda^{***} \approx 0.5392$, the root of

$$2^{-\lambda} + \tfrac19\big(\tfrac38\big)^\lambda + \tfrac29\big(\tfrac34\big)^\lambda + \tfrac19\big(\tfrac{9}{32}\big)^\lambda = 1.$$

Proof. Add to the three families of Section 5.1 a fourth: for $m' \in A \cap (x^{9/64}, x^{9/32} - 1]$, the set $\mathcal{O}(m')$; then $I(m') \subseteq (\sqrt{x}, x]$ because $m'^{32/9} > \sqrt{x}$ and $(m'+1)^{32/9} \le x$. Its members are odd with odd $\lfloor n^{3/2}\rfloor$, hence disjoint from the even members and from the OE-type members of the first three families, and distinct m' give disjoint $I(m')$. By Proposition C.4 the family has log-mass at least $\frac19(1-\varepsilon_7(x))\big(g_A(9t/32) - 2x^{-9/32}\big)$ with $\varepsilon_7 \to 0$, so (5.2) gains the term $(\frac19 - \varepsilon_7)g_A(9t/32)$. Lemma 5.1 applies with the additional pair $(e_4, c_4) = (\frac{9}{32}, \frac19)$ and the new root. □

Downstream, every constant of Sections 7–10 improves: the rate threshold of Theorems 7.2–7.3 becomes $e > 1 - \lambda^{***} \approx 0.4608$; the least depth constant of Corollary 8.4 and Theorem 9.2 is $C = 18$ ($e(18) \approx 0.480$); in the one-sided form $C(0.5) = 18$, $C(0.55) = 39$. Pairing plus $OOEEE$ sits at 0.5392, above the depth-two ceiling 0.4927, because $OOEEE$ is an extra production. A further production from the words $OOOEE$ and $OOEOE$ on even blocks would give $\lambda = 0.5561$ and $C = 18$; those fibers have length $P^{5/32}$, below the threshold of the localized triple, and Lemma 3.9's trivial bound on the kernel proof is the whole interval, so the analogue of Hypothesis L for the kernel estimate of [12] is not available (fate-contagion note ğ7.4). Hypothesis L concerns consecutive odd starts in an interval and says nothing about the odd images S_{odd}; the free term ψ_F is untouched by this appendix.

Appendix D. The finite nested productions

This appendix gives the analytic input used in (5.10). Only the fixed words $V_k = (OE)^{k-1}OEE$, $1 \le k \le 6$, are used. Constants may depend on k and on an arbitrarily small $\varepsilon > 0$; no numerical constant or estimate uniform in an unbounded depth is asserted. The classical estimates are those used in Proposition 4.4.

D.1 Exact endpoints and smooth weights

Put $F(u) = \lfloor u^{3/4}\rfloor$ and $\Phi(a) = \lceil a^{4/3}\rceil$. For integers $a < b$,

$$\{n : a \le F(n) < b\} = [\Phi(a), \Phi(b)) \cap \mathbb{Z}. \tag{D.1}$$

If n has word V_k, repeated use of Lemma 3.2 gives $J^{2i}(n) = F^i(n)$ for $0 \le i \le k$, followed by the last even step. With $w = F^{k-1}(n)$, its final state is $J^{2k+1}(n) = \lfloor w^{3/8}\rfloor$. Thus the exact landing window over $m \geq 2$ is

$$I_k(m) = [\Phi^{k-1}(a), \Phi^{k-1}(b)) \cap \mathbb{Z}, \quad a = \lceil m^{8/3} \rceil, \quad b = \lceil (m+1)^{8/3} \rceil. \tag{D.2}$$

It contains precisely the desired preimages after imposing the parity conditions of V_k. Let $s = (3/4)^{k-1}$, $\rho_k = 3s/8$, and $P = m^{1/\rho_k}$. The mean value theorem, applied successively to $u^{4/3}$, gives endpoint error $O_k(P^{1-s})$ relative to the smooth window $[m^{1/\rho_k}, (m+1)^{1/\rho_k})$. Consequently

$$|I_k(m)| = \rho_k^{-1} m^{1/\rho_k - 1}(1 + o(1)), \quad \frac{|I_k(m) \triangle I_k^{\mathrm{sm}}(m)|}{|I_k^{\mathrm{sm}}(m)|} = O_k(P^{-5s/8}). \tag{D.3}$$

Here and below interval length and integer count differ by at most a bounded endpoint error.

We also need the multiplicity of an inner-layer integer after earlier layers have been summed out. For a fixed $i \geq 1$, put $p_i = (4/3)^i$. Induction in (D.1) gives

$$\Phi^i(u) = u^{p_i} + O_i(u^{p_i - 4/3}).$$

For $i = 1$ the error is $O(1)$; each further inverse step multiplies the previous error by $O(u^{p_i/3})$, and its own rounding error is smaller. Taking the difference at $u+1$ and u, and then counting odd integers, shows that the number of odd n with $F^i(n) = u$ is

$$\omega_i(u) = \tfrac{1}{2} p_i u^{p_i - 1} + O_i(u^{p_i - 4/3} + 1) = \tfrac{1}{2} p_i u^{p_i - 1}(1 + O_i(u^{-1/3})). \tag{D.4}$$

The last form uses $p_i \geq 4/3$. This weight has a smooth monotone main term and a pointwise power-saving error. On any layer interval of scale W, partial summation transfers bounds for all initial partial sums to this main weight; the pointwise error costs $O_i(W^{-1/3})$ relative to the total mass. At most two partial end fibers occur at each grouping step. Their relative size is at most $O_k(P^{-5s/8})$, or a stronger power, since the shortest relative layer interval has length $W^{5/8}$.

D.2 Averaged cancellation inside the short fibers

Let v range over consecutive integers in an interval at scale W, of length $L \gg W^{5/8}$, contained in a fixed dyadic range. Write $B_v = [v^{4/3}, (v+1)^{4/3}) \cap \mathbb{Z}$. The following bound holds on either parity class of u, and on the full integer lattice:

$$\sum_v \left| \sum_{u \in B_v} \psi(u^{3/2}) \right| \ll L W^{1/3} W^{-1/6} (\log W)^2. \tag{D.5}$$

The same right side bounds the sum over any initial subinterval of the displayed v-range.

To see this, use Vaaler at $Q = \lfloor W^{1/6} \rfloor$. A nonzero Fourier mode on a parity class has phase $q(2r+b)^{3/2}/2$. Its derivative at the beginning of a v-fiber is $\alpha_q(v) = \frac{3}{2} q v^{2/3} + O(qW^{-2/3})$, and changes by $O(qW^{-1/3})$ across that fiber. As a function of v, the leading frequency increases in steps comparable to $\delta = qW^{-1/3}$. The full lattice has the same estimates with a different fixed leading constant.

Fibers whose derivative range meets an integer, or lies within a constant multiple of δ of one, are counted trivially. In each unit frequency interval there are $O(1)$ such fibers; there are $O(L\delta + 1)$ unit intervals. They contribute $O(qL + W^{1/3})$, since a fiber has $O(W^{1/3})$ points. For the remaining fibers, Kusmin–Landau bounds the mode by the reciprocal distance of its derivative range to the nearest integer. Grouping those distances in intervals of length comparable to δ gives a harmonic sum. Including the two incomplete unit intervals at the ends, the total for this mode is

$$\ll qL + L \log W + W^{1/3}(1 + q^{-1} \log W) \ll qL + L \log W. \tag{D.6}$$

The last step uses $L \gg W^{5/8}$. On an initial subinterval use its length in the first bound, and the full L to dominate the endpoint terms; thus the bound needed for partial summation is uniform. Summing (D.6) against coefficients $O(1/q)$ gives $O(LQ + L(\log W)^2)$. The error majorant contributes $O(LW^{1/3}/Q + LQ + L\log W)$, by its $O(1/Q)$ coefficients. These prove (D.5). Bounded coefficients depending only on v may be inserted after the inner absolute value.

In terms of the outer variable scale $Z = W^{4/3}$, the relative saving in (D.5) is $Z^{-1/8}$, up to logarithms. This averaged estimate does not assert cancellation on every individual fiber, which Lemma 4.7 shows would be false.

D.3 The final one-variable estimate

On an interval K of scale W and length $L \asymp W^{5/8}$, each nonconstant product of the three signs

$$(-1)^u, \qquad \psi(u^{3/2}), \qquad \psi(u^{3/4})$$

has sum $O_\varepsilon(LW^{-1/6+\varepsilon})$. The same majorant holds for every initial subinterval of K.

If the fast sign occurs, expand the fast and slow waves at $Q = \lfloor W^{1/6} \rfloor$. A nonzero fast mode has phase $(qu^{3/2} + ru^{3/4} + bu)/2$, with $b \in \{0,1\}$, $q \neq 0$, and $|q|, |r| \leq Q$. Its second derivative has constant sign and magnitude comparable to $|q|W^{-1/2}$. The second-derivative test bounds the sum by

$$\ll L|q|^{1/2}W^{-1/4} + |q|^{-1/2}W^{1/4}.$$

The coefficient sums, the two majorants and their bounded product give

$$\ll (LW^{-1/4}Q^{1/2} + W^{1/4} + L/Q)(\log W)^2 \ll LW^{-1/6}(\log W)^2.$$

For the slow majorant, the derivative is $\asymp |r|W^{-1/4}$, below $1/2$, so Kusmin–Landau gives the needed $O(W^{1/4}/|r|)$ mode bound.

If only the slow wave occurs, pairing consecutive level sets of $\lfloor u^{3/4} \rfloor$ gives $O(LW^{-1/4} + W^{1/4})$. With the extra sign $(-1)^u$, each level set instead has alternating sum $O(1)$, giving the same bound. The lone sign $(-1)^u$ sums to $O(1)$. All arguments retain these majorants on initial subintervals, including their partial end cells. These estimates also apply with the smooth weights (D.4).

D.4 Assembly for the six words

For $w_i = F^i(n)$, the indicator of V_k on odd starts is

$$2^{-2k}(1 + \psi(n^{3/2})) \prod_{i=1}^{k-1} (1 - (-1)^{w_i})(1 + \psi(w_i^{3/2}))(1 + \psi(w_{k-1}^{3/4})). \tag{D.7}$$

For $k = 1$ the product is empty. Every factor records the corresponding next parity after the preceding restrictions, so (D.7) is an exact indicator identity; the iterates w_i are never replaced by a single floor of a power of n.

Expand (D.7). Its constant term gives 2^{-2k} times the number of odd integers in $I_k(m)$. For each other term, select the earliest layer with a remaining sign. Earlier layers sum to the weight (D.4), including its negligible remainder. If that layer has a fast sign and is not the final layer, group by the next-layer integer and apply (D.5). An additional sign $(-1)^u$ is handled by splitting into its two parity classes. All later-layer factors are constant on the grouped fiber and have magnitude at most one. If that layer has only $(-1)^u$, sum alternately within each next-layer fiber; partial summation gives an error at most a constant times the smooth weight per fiber, saving a factor $Z^{-1/4}$ at layer scale Z. If no earlier sign remains, apply Section D.3 on the final layer.

The scale of layer i is $Z_i = P^{(3/4)^i}$, and its interval length has order $Z_i^{1-\rho_k/(3/4)^i}$. The final layer has relative length $5/8$; every preceding layer has a longer relative interval. Thus all applications above meet their length condition. The weakest fast-layer saving is $Z_{k-2}^{-1/8} = P^{-s/6}$, and the final-layer estimate saves $Z_{k-1}^{-1/6} = P^{-s/6}$. For $k = 1$ only Section D.3 is needed. The weight and endpoint errors in D.1 are smaller. There are finitely many terms for each $k \le 6$, so

$$\#\{n \in I_k(m) : \mathrm{word}_{2k+1}(n) = V_k\} = 2^{-2k}\#\{n \in I_k(m) : n \text{ odd}\} + O_{k,\varepsilon}(|I_k(m)|P^{-s/6+\varepsilon}). \tag{D.8}$$

The location and length of the fiber, together with $s/(6\rho_k) = 4/9$, turn this into the two-sided logarithmic mass estimate

$$\sum_{\substack{n \in I_k(m) \\ \mathrm{word}_{2k+1}(n) = V_k}} \frac{1}{n} = \frac{3^{-k}}{m}\left(1 + O_{k,\varepsilon}(m^{-4/9+\varepsilon})\right). \tag{D.9}$$

For shell sums, discard the bounded number of terminal source integers whose exact fiber meets a shell boundary. The upper endpoint perturbation in (D.3) is smaller than a source step at that endpoint, and the lower endpoint is treated identically. Their reciprocal source mass is exponentially small in the shell logarithm. Summing (D.9) therefore gives the claimed coefficient 3^{-k} at source scale $\rho_k t$, with vanishing errors. Finally the prefix-free words give disjoint source sets, and the subtraction in (5.9) proves (5.10). This completes the analytic input to Theorem 5.3. □

Acknowledgments and use of AI

Large language models assisted the development of this work, including prose, proposed proof arguments, Lean formalizations, and computations. The September 2026 review with OpenAI Codex revised the analytic estimates and reductions, checked references, ran the numerical and formal audits, and prepared this preprint. AI assistance is not independent mathematical validation. The author is responsible for the statements, proofs, code, and the decision to make this version public. The formalization covers only the results identified in Section 1.4 and Appendix A.

Availability and version

This version is prepared as a Zenodo preprint. The accompanying source package contains the Markdown and LaTeX sources, figures, the local Lean dependency closure with its pinned toolchain, numerical audit code, archived experiment records, and a manifest of file hashes. The repository is sneakyweasel/btlab. The original development notes remain historical records; where their statements differ, the corrected proofs and scopes in this version of Paper C govern this preprint.

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